

Studies of Tracking and Reconstruction in ATLAS: GPU-based Strip Clustering and
Optimization of a Run 3 Search for Higgs Decays to Dark Photons

by

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DISSERTATION ABSTRACT

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Master of Science in Physics

Title: Studies of Tracking and Reconstruction in ATLAS: GPU-based Strip Clustering and Optimization of a Run 3 Search for Higgs Decays to Dark Photons

This thesis presents two independent studies: a GPU-based strip-clustering algorithm designed for the high-luminosity environment of the ATLAS experiment at the Large Hadron Collider (LHC) and an optimization of an ATLAS Run 3 search for Higgs boson decays to dark photons.

The first study focuses on the development and validation strip-detector clustering algorithm in **Traccc**, a GPU-accelerated tracking toolkit (usable for both offline track reconstruction and the online trigger system) designed to outperform traditional CPU-based algorithms. To support more data processing in a higher-collision-per-second environment during the Phase-II upgrade program for the High-Luminosity LHC (HL-LHC), we propose Event Filter (EF) Tracking GPU pipelines that use GPU-based algorithms. The strip clusterization algorithm groups cell hits from silicon strip sensors to form measurements and is integrated into the EF Tracking GPU pipelines in the **ATHENA** framework. The algorithm is validated and shows good consistency in comparing differences in local measurement positions between **TRACCC** and **ATHENA**.

Alongside this detector-focused work, there is a contribution to a search for Higgs boson decays to photons + dark photons produced via gluon-gluon fusion. The search is based on an integrated luminosity of 135 fb^{-1} of ATLAS Run3 data; we also provide a projection to the full Run 3 dataset of $xx \text{ fb}^{-1}$. Motivated by

finding a potential candidate for dark matter, we search for events with the final state of a Standard Model (SM) photon, missing transverse energy (E_T^{miss}), and collimated sprays of hadrons called jets. My analysis aims to maximize the signal-to-background significance by optimizing selection criteria based on distributions of various variables. A Machine Learning (ML) classifier study, including Boosted Decision Tree (BDT) and Neural Network (NN) approaches, was also developed to further enhance signal sensitivity. After optimizing the selection criteria, the analysis improved the expected upper limit on the branching ratio of this process compared to the previous work by 20% (from 1.1 to 0.88). When projected to the full Run 3 dataset, the expected upper limit improved to xx.

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To so-and-so...

TABLE OF CONTENTS

Chapter	Page
I. INTRODUCTION	16
1.1. The Standard Model of particle physics	16
1.1.1. Brief Overview of SM particles and interactions	16
1.1.2. Role of the Higgs boson	19
1.2. Beyond the Standard Model	23
1.3. ATLAS Run 3 Detector Configuration	25
1.3.1. Introduction	25
1.3.2. Timeline of LHC and ATLAS Data-Taking Runs	26
1.3.2.1. Run 1 (2010–2012)	26
1.3.2.2. Run 2 (2015–2018)	26
1.3.2.3. Run 3 (2022–2026)	27
1.3.2.4. Long Shutdown 3 and Run 4 (HL-LHC Era)	27
1.3.3. The ATLAS Experimental Complex	28
1.3.3.1. Luminosity	29
1.3.3.2. Pile-up	30
1.3.4. The ATLAS Detector	32
1.3.4.1. Inner Detector	32
1.3.4.2. Calorimeter System	33
1.3.4.3. Muon Spectrometer	33
1.3.5. Trigger and Data Acquisition System	34
1.4. ATLAS Run 4 Upgrade	35

Chapter	Page
II. ATLAS EVENT FILTER (EF) TRACKING: STRIP CLUSTERING	37
2.1. Introduction	37
2.1.1. Overview of GPU-based pipelines	38
2.2. Overview of GPU-based pipelines	40
2.2.0.1. Motivation for GPU Implementation	42
2.2.1. Previous work and motivation	43
2.2.2. Overview of GPU-based Pipelines	43
2.2.3. Traccc and Athena framework	43
2.2.4. Strip clustering algorithm	43
2.2.5. Input data	44
2.2.6. Integration with the Traccc framework	45
2.3. Results and Discussion	45
2.3.1. Cluster comparison with CPU	45
2.3.2. Efficiency	48
2.3.3. Summary of EF Tracking validation	48
III. ATLAS: SIGNAL OPTIMIZATION OF A SEARCH FOR HIGGS DECAYS TO DARK PHOTONS USING ATLAS RUN 3 DATA	49
3.1. Introduction	49
3.1.1. Motivation for Hidden Sectors	49
3.1.2. Kinetic Mixing and the Dark Photon	49
3.2. Data and MC samples	52
3.2.1. Data samples	52
3.2.2. MC samples	53
3.3. Physics object reconstruction and identification	54

Chapter	Page
3.4. Event selection	58
3.4.1. Trigger	58
3.5. Signal region optimization	58
3.5.1. N-1 iteration method	59
3.5.2. Statistical method	61
3.5.3. Machine Learning approach	69
3.5.3.1. Boosted Decision Tree (BDT) training and results	69
3.5.3.2. TRExFitter for BDT	74
3.5.3.3. Neural Network (NN) training and results	74
3.6. Conclusions and Outlook	78

APPENDICES

A. DARK PHOTON: DATA AND MC SAMPLES	79
B. IRIS-HEP FELLOWSHIP PROJECT: ANALYTICAL TRACKING VALIDATION	82
B.1. Overview of the Project	82
B.1.1. Motivation and Goals	82
B.1.2. A Common Tracking Software (ACTS) framework and Open Data Detector (ODD) Geometry	82
B.1.3. Python Tracking Resolution Calculator	82
B.2. Methodology	82
B.2.1. Analytical Resolution Formulation	82
B.2.2. Measurement and Multiple Scattering Terms	82
B.2.3. Matrix Representation of Track Parameters	83
B.3. ACTS Configuration and Simulation Setup	83

Chapter	Page
B.4. Comparison Strategy	83
B.5. Results and Discussion	84
B.5.1. Resolution vs. p_T and η	84
B.5.2. Discrepancies and Model Validation	84
B.5.3. Implications for Fast Performance Estimation	85
B.6. Conclusions	85
B.6.1. Summary of Findings	85
B.6.2. Cross-Project Insights	85
B.6.3. Future Work	85
B.7. Supplementary Material	85
B.7.1. ACTS and Traccc Configuration Files	85
B.7.2. Code Snippets and YAML Settings	85
B.7.3. Additional Figures	85
REFERENCES CITED	86

LIST OF FIGURES

Figure		Page
1.	Elementary particles of the Standard Model [2].	17
2.	Feynman diagrams contributing to the Higgs production in (a) gluon fusion, (b) weak-boson fusion, (c) Higgs-strahlung (or associated production with a gauge boson), and (d) associated production with top quarks [?].	21
3.	Leading-order contributions to the decay of the Higgs boson into two photons [6].	22
4.	Schematic of CERN accelerator complex [19]	29
5.	Cumulative integrated luminosity recorded by ATLAS during Run 3 [20]	30
6.	Vertex topologies in ATLAS, including primary, secondary and pile-up vertices	31
7.	ATLAS detector slice with labeled subdetectors and particle visualizations [21]	32
8.	ATLAS Trigger and Data Acquisition System in Run 3 [22]	34
9.	Overview of the reconstruction pipelines, including the G-050, G-130, and G-200 pipelines discussed in this thesis.	42
10.	Validation of strip cluster local coordinates and η dependence.	46
11.	Global coordinate distributions for the strip clusters.	47
12.	Feynman diagram illustrating the Higgs boson production via gluon-gluon fusion process (top) and vector-boson fusion (VBF) and Higgsstrahlung (VH) processes (bottom left and right), all followed by the decay into a photon and a dark photon ($H \rightarrow \gamma\gamma_d$). The hatched circle in the $H\gamma\gamma_d$ vertex represents the interaction through messenger fields.	51
13.	Iteration workflow for optimizing the cut set using n-1 studies and significance checks.	59

Figure	Page
14. Generated by Tim Matthew (senior PhD at University of Oregon), pre-fit (top row) and post-fit (bottom row) m_T distributions in the Signal Region (SR), 1μ Control Region, and 2μ Control Region. The distributions span 80–300 GeV with binning (80, 110, 140, 200, 300) GeV.	63
15. Normalization factors for W and Z processes and signal strength, with related statistical uncertainties	63
16. Observed and expected 95% CL upper limits on the branching ratio $\mathcal{B}(H_{125} \rightarrow \gamma\gamma_D)$ obtained from the individual analysis channels and their combination. The black markers indicate the observed limits, while the dashed line represents the median expected limit under the background-only hypothesis. The green and yellow bands correspond to the $\pm 1\sigma$ and $\pm 2\sigma$ intervals on the expected limit, respectively [64].	64
17. Additional Selection variables distributions after the optimized rectangular cuts for the ggF $H \rightarrow \gamma\gamma_D$ analysis.	67
18. N–1 distributions for the ggF $H \rightarrow \gamma\gamma_D$ analysis.	68
19. Correlation matrix of the BDT input variables after the baseline selection. The absolute value of the correlation above 0.75 will drop one of the two variables based on their physical significance and their AUC of ROC.	70
20. Weight and Biases plot [68] for the 95 BDT Sweeps (null runs were excluded), where it records the hyperparameters (gamma, learning rate, max depth, number of estimators, subsample) and performance (test AUC of the ROC, best $\frac{S}{\sqrt{B}}$, its corresponding ML BDT Score cut and acceptance rate). The highlighted line is the chosen parameter in table 10.	72
21. Significance versus BDT score threshold, with the maximum simple significance at BDT score ≥ 0.02 . $S/\sqrt{B}$ (blue) is the simple counting significance; $S/\sqrt{S+B}$ is (orange) the poisson-aware approximation; $S/\sqrt{S+1.3B}$ (green) is the Ad-hoc systematic-aware metric, assuming 30% background systematic; Zbi (red) is the Binomial significance.	73
22. Performance of the BDT classifier. (Left) Receiver operating characteristic (ROC) curve. (Right) Feature importance of input variables.	74

Figure	Page
23. Significance versus NN score threshold. The maximum simple significance ($S/\sqrt{B}$ = 3.052) is achieved at NN score $\geq$ 0.03, corresponding to a signal acceptance of 64.04%.	77

LIST OF TABLES

Table	Page
1. Average timing of the different steps of track reconstruction using the G-O80 pipeline for $t\bar{t}$ events at $\langle\mu\rangle = 140$ and $\langle\mu\rangle = 200$	45
2. Summary of the Good Run Lists (GRL) and integrated luminosities used for the Run 3 analysis.	52
3. Electron selection criteria.	55
4. Jet reconstruction criteria.	55
5. Muon selection criteria.	56
6. Overlap removal criteria.	56
7. Photon selection criteria.	57
8. Triggers used in the analysis.	58
9. Preselections and basic-cut comparison between the Internal Note and Optimized Selections	61
10. Hyperparameter distribution in the randomized BDT sweep and the parameters for the best run	71
11. Model of the Deep Neural Network (DNN) Classifier	75
12. Hyperparameter search ranges used in the neural network Bayesian sweep and the best-performing configuration.	76
13. Comparison of the four methods with their performance in observed and expected limit, significance, and significance x acceptance	78
A.14 List of signal MCd samples used in this analysis.	79
A.15 List of $\gamma jets$ MC23d and MC23e samples used in this analysis.	79
A.16 List of $W\gamma/Z\gamma$ MC23d and MC23e samples used in this analysis	80
A.17 List of W/Z MC23d and MC23e samples used in this analysis.	81

CHAPTER I

INTRODUCTION

1.1 The Standard Model of particle physics

1.1.1 Brief Overview of SM particles and interactions. The Standard Model of particle physics is a relativistic quantum field theory that describes the elementary particles known to date that constitute matter and the four fundamental forces of nature (except gravity). At the Large Hadron Collider (LHC) in 2012, the latest particle discovery was the Higgs boson, which is responsible for electroweak symmetry breaking and mass generation [1].

This chapter introduces the components and their significance in the Standard Model, while the next section, beyond the Standard Model, explains the limitations of the current model.

The Standard Model is based on the quantum field theory described by the symmetry group of

$$SU(3)_C \times SU(2)_L \times U(1)_Y.$$

Particles in the Standard Model are categorized into fermions and bosons. Each particle has its own antiparticle with opposite charges, while all other properties remain the same [3] [4].

Fermions are building blocks of matter that have spin $\frac{1}{2}$ and consist of three generations of quarks and leptons. These particles obey Fermi-Dirac statistics and are subject to the Pauli exclusion principle, which states that no two fermions can occupy the same quantum state. It explains that electrons move to higher-energy states when lower-energy states are occupied, thereby forming the atomic structure we know today. The six flavors of quarks are up, down, charm, strange, top, and bottom quarks. Organized by generations, the increasing mass from left to right,

Standard Model of Elementary Particles

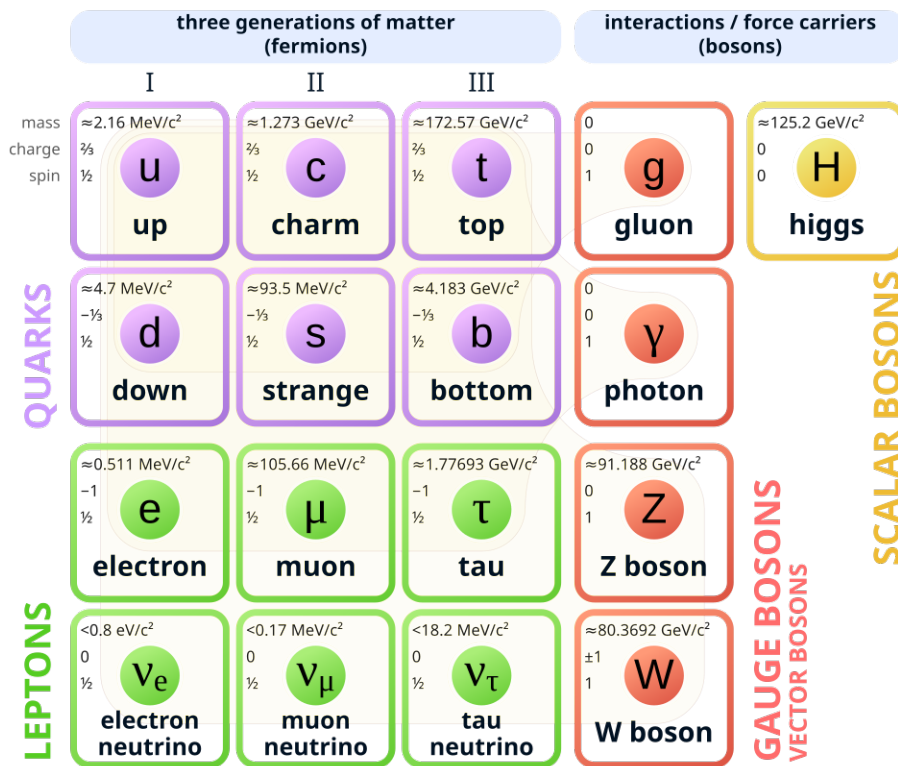


Figure 1. Elementary particles of the Standard Model [2].

the quarks are represented as:

$$\begin{pmatrix} u \\ d \end{pmatrix}, \begin{pmatrix} c \\ s \end{pmatrix}, \begin{pmatrix} t \\ b \end{pmatrix}$$

The first row of the quarks possesses a fractional charge of $+\frac{2}{3}$; the second row has $-\frac{1}{3}$. Their corresponding antiparticles are

$$\begin{pmatrix} \bar{u} \\ \bar{d} \end{pmatrix}, \begin{pmatrix} \bar{c} \\ \bar{s} \end{pmatrix}, \begin{pmatrix} \bar{t} \\ \bar{b} \end{pmatrix}$$

All these quarks have a non-Abelian charge called color charge from the local gauge symmetry $SU(3)_c$ of quantum chromodynamics (QCD). Due to color confinement, they cannot exist by themselves; instead, they appear as color-neutral particles,

known as hadrons. Hadrons are further categorized into mesons (quark-antiquark) and baryons (three quarks or three antiquarks). Mesons consist of a quark-antiquark pair, like the triplets of pions (π^+ , π^0 , π^-), where the color cancels out a color from the quark and its corresponding anti-color from the antiquark. Examples of baryons are protons (p) and neutrons (n), where color neutrality is from the balance of the three colors of each quark, commonly denoted as red, green, and blue. Quarks interact through all four fundamental forces. The strong force, mediated by gluons, binds quarks into hadrons; the electromagnetic and weak interactions are responsible for charge and flavor changes; the gravitational force, due to its small magnitude, is irrelevant compared to the other three forces.

The leptons are the electron, muon, tau, electron neutrino, muon neutrino, and tau neutrino. Organized by generations and increasing mass from left to right, the leptons are:

$$\begin{pmatrix} e^- \\ \nu_e \end{pmatrix}, \begin{pmatrix} \mu^- \\ \nu_\mu \end{pmatrix}, \begin{pmatrix} \tau^- \\ \nu_\tau \end{pmatrix}.$$

The first row of leptons has a charge of -1, and neutrinos are charge-neutral and left-handed. Their corresponding anti-particles are

$$\begin{pmatrix} e^+ \\ \bar{\nu}_e \end{pmatrix}, \begin{pmatrix} \mu^+ \\ \bar{\nu}_\mu \end{pmatrix}, \begin{pmatrix} \tau^+ \\ \bar{\nu}_\tau \end{pmatrix}.$$

The first row of anti-leptons becomes +1 charge, and anti-neutrinos are right-handed.

Charged leptons interact via the electromagnetic, weak, and gravitational interactions, whereas neutrinos interact only via the weak interaction and gravity, making them weakly interacting and difficult to detect.

Lepton flavor is conserved in electromagnetic and strong processes, but can change through weak interactions, a phenomenon most strikingly observed in neutrino oscillations, which demonstrate that neutrinos have nonzero but tiny masses. The increasing masses across generations, particularly the large tau mass relative to the electron, reflect a hierarchical structure whose origin remains unexplained within the Standard Model.

Bosons comprise four spin-1 gauge bosons and one spin-0 Higgs Boson. The gauge bosons are fundamental force mediators: the photon for electromagnetism, the W and Z bosons for weak interaction, and the gluons for the strong interaction. These particles mediate forces via quantum field interactions and are responsible for binding charged particles, inducing particle decays, and confining quarks within hadrons. Notice that there is no gauge boson for gravitation. Although theory predicts a massless spin-2 graviton as the force carrier, it has not yet been experimentally verified. In addition to gauge bosons, the Standard Model contains the Higgs boson, a scalar (spin-0) particle associated with the Higgs field, whose nonzero vacuum expectation value gives rise to the masses of elementary fermions and weak gauge bosons through spontaneous electroweak symmetry breaking.

1.1.2 Role of the Higgs boson. The Higgs boson is a fundamental scalar particle associated with the Higgs field, which is introduced in the Standard Model to explain the origin of mass for elementary particles while preserving gauge invariance [4]. The Higgs field is a complex scalar doublet transforming under the electroweak gauge group $SU(2)_L \times U(1)_Y$, and its dynamics are governed by the Higgs potential

$$V(\Phi) = \mu^2 \Phi^\dagger \Phi + \lambda (\Phi^\dagger \Phi)^2, \quad (1.1)$$

where $\mu^2 < 0$ and $\lambda > 0$. The negative quadratic term leads to spontaneous symmetry breaking, resulting in the Higgs field acquiring a nonzero vacuum expectation value.

Minimizing the potential yields

$$\langle \Phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \quad (1.2)$$

where $v = \sqrt{-\mu^2/\lambda} \approx 246$ GeV is the electroweak vacuum expectation value. This vacuum configuration breaks the electroweak symmetry down to the electromagnetic subgroup $U(1)_{\text{EM}}$.

As a consequence of electroweak symmetry breaking, the weak gauge bosons acquire mass through their interactions with the Higgs field. The resulting masses are given by

$$m_W = \frac{1}{2}gv, \quad m_Z = \frac{1}{2}\sqrt{g^2 + g'^2}v, \quad (1.3)$$

while the photon remains massless. Fermions acquire mass through Yukawa interactions of the form

$$\mathcal{L}_{\text{Yukawa}} = -y_f \bar{\psi}_L \Phi \psi_R + \text{h.c.}, \quad (1.4)$$

leading to fermion masses

$$m_f = \frac{y_f v}{\sqrt{2}}, \quad (1.5)$$

where y_f denotes the Yukawa coupling of fermion f .

The Higgs boson itself arises as a physical excitation around the vacuum expectation value of the Higgs field. Its mass is determined by the curvature of the Higgs potential at the minimum,

$$m_H = \sqrt{2\lambda}v, \quad (1.6)$$

and has been extensively measured experimentally by ATLAS and CMS at the LHC to be

$$m_H = 125.20 \pm 0.11 \text{ GeV} \quad (1.7)$$

as reported by the Particle Data Group [5]. Unlike gauge bosons, the Higgs boson does not mediate a force, but instead plays a central structural role by setting the mass scale of the Standard Model. Its couplings are proportional to particle masses, making it a unique probe of the electroweak symmetry-breaking mechanism and a sensitive window into possible physics beyond the Standard Model.

At the LHC, the dominant production mode for Higgs bosons is gluon fusion ($gg \rightarrow H$), mediated by a top quark loop. Additional production mechanisms include vector boson fusion (VBF), associated production with a vector boson (WH or ZH), and production with top quarks ($t\bar{t}H$). The Feynman diagrams are shown in Fig. 3.

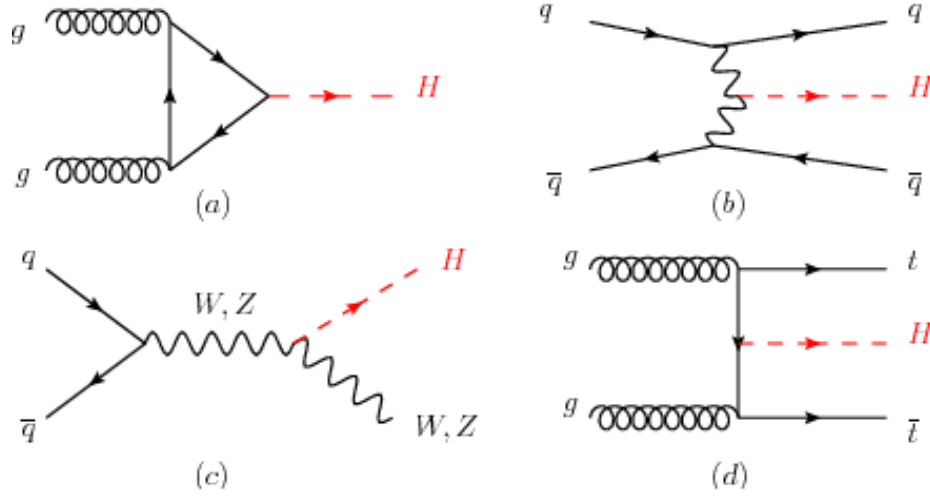


Figure 2. Feynman diagrams contributing to the Higgs production in (a) gluon fusion, (b) weak-boson fusion, (c) Higgs-strahlung (or associated production with a gauge boson), and (d) associated production with top quarks [?].

Once produced, the Higgs boson decays into several Standard Model final states. The dominant decay channel is $H \rightarrow b\bar{b}$, while other important channels include $H \rightarrow WW^*$, $H \rightarrow ZZ^*$, $H \rightarrow \tau^+\tau^-$, and the loop-induced decay $H \rightarrow \gamma\gamma$. A Feynman diagram of the diphoton decay channel is shown in Fig. 3.

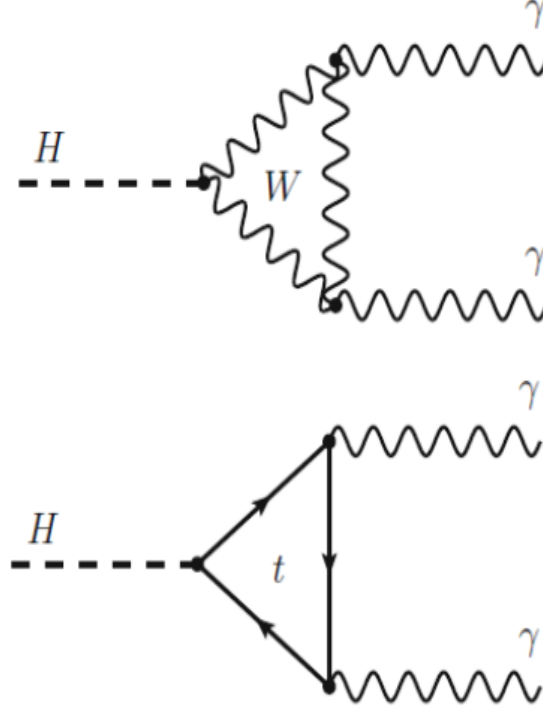


Figure 3. Leading-order contributions to the decay of the Higgs boson into two photons [6].

Because the Higgs boson couples to heavy particles and can interact with new states through loop processes, it provides a sensitive probe of physics beyond the Standard Model. For example, models with introducing the dark sector (γ_D) allow the decay $H \rightarrow \gamma\gamma_D$, where the dark photon interacts with the Standard Model through kinetic mixing. This process produces a final state containing a

photon and missing transverse momentum, leading to the dark photon analysis in Section ??.

1.2 Beyond the Standard Model

The Standard Model provides a successful description of the known elementary particles and their interactions. Various experimental measurements confirm the model's predictions. However, the model has limitations that cannot explain some experimental observations, prompting researchers to search for Beyond the Standard Model (BSM) physics [5] [4].

As briefly mentioned in Subsection 1.1.1, electromagnetic, weak, and strong interactions, with their corresponding mediator particles, are incorporated into the quantum field theoretical framework, whereas gravity is currently understood only through classical mechanics and general relativity. The theory hypothesizes a force carrier for gravity, graviton (a massless, neutral-charged, spin-2 boson), whereas it has not been experimentally observed. Mathematically, the theory of quantum gravity treats the force carrier as a point particle, leading to a non-renormalizable infinite result when calculating the graviton-graviton interaction.

The Standard Model also fails to account for the existence of dark matter, which makes up around 26.8% of the whole universe [7]. Astrophysical and cosmological observations, including galaxy rotation curves [8], gravitational lensing [9], and measurements of the cosmic microwave background [7], provide compelling evidence for a non-luminous form of matter that constitutes a significant fraction of the energy density of the universe. However, none of the Standard Model particles satisfy the properties of dark matter. The latter

Section ?? proposes one of the potential solutions: the Hidden Sector hypothesis, where the dark photons can interact with SM photons through kinetic mixing.

There is also the flavor problem that remains unexplained, referring to the three generations, mass hierarchy, and the mixing angles. The Standard Model organized the particles into three generations, yet there is no fundamental justification for this. Secondly, the masses of the particles are determined by the Yukawa coupling constants to the Higgs field, which are experimentally determined. The mass of the electron is approximately $0.511 \text{ MeV}/c^2$, whereas the top quark mass is about $(172.56 \pm 0.32) \text{ GeV}/c^2$ [5]. Such a drastic mass difference means there is no symmetry or principle to predict their coupling strength. Moreover, particles can transit from one flavor to another. Quark mixing is governed by the Cabibbo-Kobayashi-Maskawa (CKM) matrix, which connects their weak and mass eigenstates. Within this framework, the mixing angles are relatively small, indicating the quarks' tendency to stay within their generations. Nevertheless, neutrino mixing is described by the Pontecorvo-Maki-Nakagawa-Sakata (PMNS) matrix, in which the mixing angles are relatively large. As a result, the neutrinos are in flavor states that are a superposition of the mass states (ν_1, ν_2, ν_3) . The difference in mass causes a discrepancy in travel speed and phases, forming a new combination that resembles a muon neutrino rather than an electron neutrino, for example. This is the neutrino oscillation.

Another unresolved issue is the neutrino masses. The Standard Model predicts that the neutrinos are exactly massless due to the absence of right-handed neutrino fields and Yukawa couplings. Nonetheless, experimental observations of neutrino oscillations demonstrate nonzero masses and flavor-state interconversions [10] [11]. Explaining these masses requires introducing new

particles, new interactions, or higher-dimensional operators beyond the Standard Model.

In addition, the Standard Model does not explain the observed matter–antimatter asymmetry of the universe [12]. Although it contains sources of CP violation in the quark sector, these effects are insufficient to account for the baryon asymmetry generated in the early universe. Additional sources of CP violation or departures from thermal equilibrium are therefore required.

Taken together, these open questions provide strong motivation for the study of physics beyond the Standard Model. Addressing these issues requires extending the particle content, symmetries, or interactions of the theory, often leading to new degrees of freedom with weak or indirect couplings to Standard Model particles. Such extensions offer concrete experimental signatures that can be tested at collider experiments and in precision measurements.

1.3 ATLAS Run 3 Detector Configuration

1.3.1 Introduction. The A Toroidal LHC ApparatuS (ATLAS) detector is a general-purpose particle physics experiment located at Interaction Point 1 (IP1) of the Large Hadron Collider (LHC) [13]. Designed to exploit the full discovery potential of the LHC, ATLAS is capable of detecting a wide range of physics signatures, from high-precision Standard Model measurements to searches for new particles such as heavy bosons, microscopic black holes, or dark matter candidates.

The detector is forward-backward symmetric with respect to the interaction point. It measures approximately 46 meters in length, 25 meters in diameter, and weighs roughly 7,000 tonnes. ATLAS is composed of concentric layers of sub-

detectors, each designed to measure specific properties of particles produced in proton-proton collisions.

1.3.2 Timeline of LHC and ATLAS Data-Taking Runs. The physics program of the ATLAS experiment is organized into successive operational periods of the Large Hadron Collider (LHC), known as runs, which are separated by long shutdowns dedicated to accelerator consolidation and detector upgrades. These runs—Run 1, Run 2, Run 3, and the coming Run 4—correspond to distinct experimental eras with progressively increasing collision energy, luminosity, and detector performance requirements.

1.3.2.1 Run 1 (2010–2012). Run 1 marked the first physics operation of the LHC, delivering proton–proton collisions at center-of-mass energies of $\sqrt{s} = 7$ TeV (2010–2011) and 8 TeV (2012). During this period, the ATLAS detector was fully commissioned, and baseline performance for tracking, calorimetry, muon reconstruction, and triggering was established.

The most significant scientific achievement of Run 1 was the discovery of the Higgs boson in 2012 by the ATLAS and CMS collaborations. From an experimental perspective, Run 1 validated the overall detector design while highlighting limitations in radiation tolerance, trigger bandwidth, and tracking performance that informed later upgrades.

1.3.2.2 Run 2 (2015–2018). Run 2 saw the LHC operate at a center-of-mass energy of $\sqrt{s} = 13$ TeV, nearly doubling the energy available in Run 1. This enabled precision measurements of Higgs boson properties, detailed

studies of top-quark and electroweak processes, and extensive searches for physics beyond the Standard Model.

A major ATLAS upgrade for Run 2 was the installation of the Insertable B-Layer (IBL), which significantly improved vertex reconstruction and impact-parameter resolution. Increased pile-up and higher trigger rates motivated substantial advances in reconstruction software and trigger strategies.

—

1.3.2.3 Run 3 (2022–2026). Run 3 operates at a center-of-mass energy of $\sqrt{s} = 13.6$ TeV, the highest achieved at the LHC to date [14]. Although the energy increase relative to Run 2 is modest, Run 3 is characterized by higher luminosity and more challenging pile-up conditions.

This run serves as a critical transition period toward the High-Luminosity LHC (HL-LHC). Upgraded trigger architectures, real-time reconstruction techniques, and improved tracking algorithms are validated under realistic conditions, preparing for Run 4 [15].

—

1.3.2.4 Long Shutdown 3 and Run 4 (HL-LHC Era).

Long Shutdown 3 (LS3) will enable the transformation of the LHC into the High-Luminosity LHC, with the goal of delivering an integrated luminosity of approximately 3–4 ab^{-1} [16]. ATLAS will undergo major Phase-II upgrades, including the complete replacement of the Inner Detector with the all-silicon Inner Tracker (ITk) [17].

Run 4 will define the HL-LHC era, characterized by unprecedented pile-up levels and data volumes. These conditions require fundamentally new approaches to tracking, triggering, and data processing, including increased parallelization and

hardware acceleration. Physics goals include percent-level measurements of the Higgs coupling, rare Standard Model processes, and enhanced sensitivity to new physics phenomena.

1.3.3 The ATLAS Experimental Complex. The ATLAS experiment is one of the four major detectors located at interaction points along the Large Hadron Collider (LHC) ring at CERN, as seen in Fig. 4. It is designed as a general-purpose detector capable of recording the products of high-energy proton–proton collisions over a wide range of final states [18]. The detector surrounds the interaction point and provides sufficient angular coverage for particle reconstruction. For this coverage, ATLAS uses a right-handed coordinate system with its origin at the nominal interaction point (IP) in the center of the detector and the z-axis along the beam pipe. The x-axis points from the IP to the center of the LHC ring, and the y-axis points upwards. Cylindrical coordinates (r, ϕ) are used in the transverse plane, ϕ being the azimuthal angle around the z -axis. The pseudorapidity is defined in terms of the polar angle θ as $\eta = -\ln \tan(\theta/2)$.

The overall experimental setup integrates the detector with the LHC accelerator infrastructure, allowing proton beams circulating in opposite directions to collide at the center of ATLAS. The detector layout is optimized to measure charged-particle trajectories, electromagnetic and hadronic energy deposits, and muon momenta, enabling precise reconstruction of collision events. This integrated design supports stable detector operation during extended data-taking periods and accommodates progressively higher collision rates across successive LHC runs.

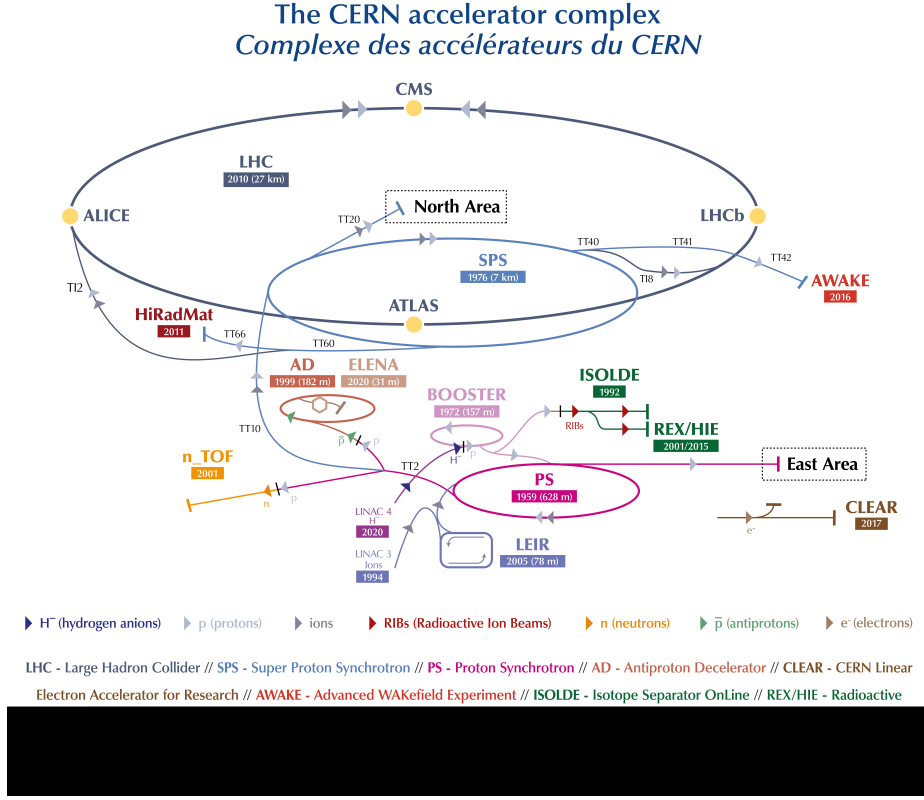


Figure 4. Schematic of CERN accelerator complex [19]

1.3.3.1 Luminosity. The physics reach of the ATLAS experiment is strongly influenced by the LHC's delivered luminosity. Luminosity characterizes the rate at which proton–proton collisions occur and determines the expected number of events for a given process with a known cross section. It is defined as

$$\mathcal{L} = \frac{N}{\sigma}, \quad (1.8)$$

where N is the number of observed events and σ is the corresponding production cross section.

The instantaneous luminosity depends on accelerator parameters such as the number of protons per bunch, the number of colliding bunches, and the transverse beam profiles at the interaction point. The total dataset collected by ATLAS is

described by the integrated luminosity, obtained by integrating the instantaneous luminosity over time, shown in Fig. 5. Integrated luminosity is typically expressed in inverse femtobarns (fb^{-1}) and represents the cumulative data available for physics analyses.

As the LHC has progressed through successive operational periods, improvements in accelerator performance have resulted in significant increases in delivered luminosity. While higher luminosity enhances sensitivity to rare processes, it also increases detector occupancy and makes reconstruction more challenging, thereby complicating the extraction of physics signals from background noise.

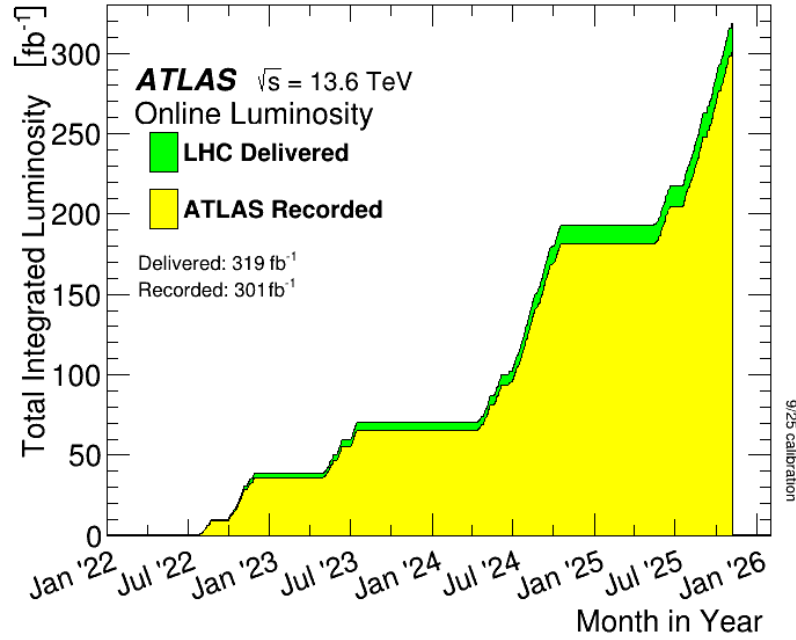


Figure 5. Cumulative integrated luminosity recorded by ATLAS during Run 3 [20]

1.3.3.2 Pile-up. Pile-up refers to multiple proton-proton interactions occurring within a single bunch crossing. It is commonly quantified by the parameter μ , which represents the average number of inelastic interactions per

bunch crossing. Higher luminosity operation of the LHC directly leads to increased pile-up.

Pile-up interactions contribute additional charged particles and energy deposits that are unrelated to the primary hard-scatter process and to the secondary vertex from b -quark decays. These additional interactions can affect track reconstruction, vertex identification, jet energy measurements, and missing transverse momentum reconstruction. As a result, pile-up represents a major experimental challenge, particularly in later LHC runs where luminosity is substantially higher.

To mitigate the impact of pile-up, ATLAS employs a combination of detector design features and reconstruction techniques, including precise vertex reconstruction and pile-up-aware object definitions. Accurate modeling of pile-up in Monte Carlo simulations is also essential, requiring simulated events to be reweighted to reproduce the pile-up conditions observed in data.

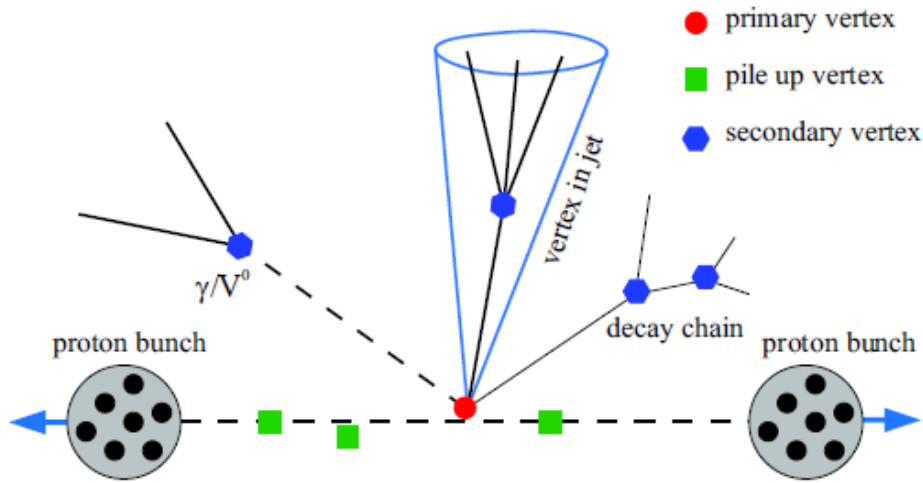


Figure 6. Vertex topologies in ATLAS, including primary, secondary and pile-up vertices

1.3.4 The ATLAS Detector.

1.3.4.1 Inner Detector. The Inner Detector is the innermost tracking system in ATLAS and reconstructs the trajectories of charged particles produced in proton–proton collisions, which consists of pixel and strip detectors and Transition Radiation Tracker (TRT, which will be removed before Run 4). It operates in a uniform solenoidal magnetic field, enabling precise measurements of charged-particle momenta via curvature. The Inner Detector provides accurate measurements of track parameters, including transverse momentum, impact parameters, and vertex positions.

By combining information from multiple tracking layers, the Inner Detector achieves high efficiency for charged-particle reconstruction over a wide pseudorapidity range. Precise vertex reconstruction enables the identification of primary interaction vertices and the separation of tracks originating from pile-up interactions. The Inner Detector plays a central role in object identification, including electrons, charged hadrons, and tracks associated with jets.

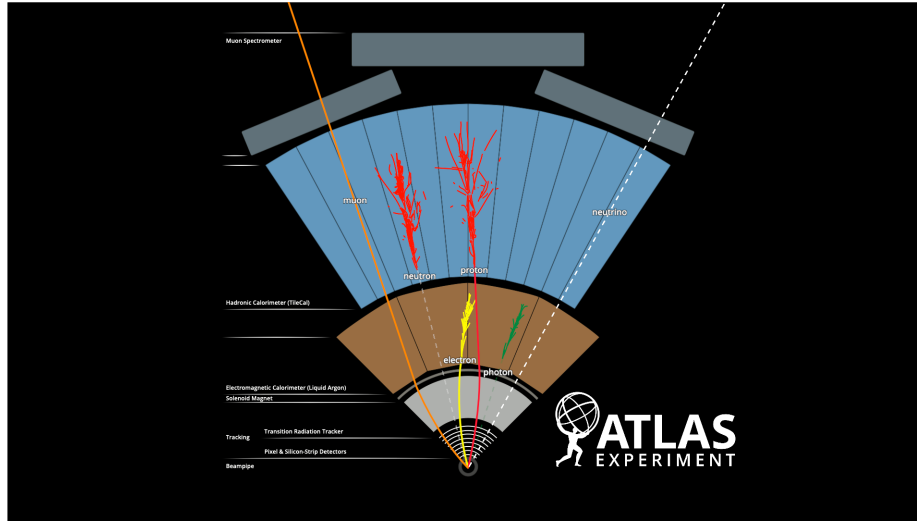


Figure 7. ATLAS detector slice with labeled subdetectors and particle visualizations [21]

1.3.4.2 Calorimeter System. The calorimeter system surrounds the

Inner Detector and is designed to measure particle energies through interactions with dense absorber materials. It is divided into electromagnetic and hadronic calorimeters, each optimized for different particle types. The electromagnetic calorimeter provides high-resolution energy measurements for electrons and photons, while the hadronic calorimeter measures the energies of hadrons and jets.

The calorimeter system provides nearly full angular coverage and is essential for jet reconstruction, electron and photon identification, and missing transverse momentum determination. In addition to energy measurements, calorimeter information is used for particle identification and plays a key role in the trigger system.

1.3.4.3 Muon Spectrometer. The Muon Spectrometer forms

the outermost detector system of ATLAS and is dedicated to the identification and momentum measurement of muons. Muons traverse the inner detector and calorimeters with minimal energy loss, allowing them to be measured in a large volume surrounding the calorimeter system.

The Muon Spectrometer employs large magnetic fields and multiple detector technologies to independently reconstruct muon trajectories. Momentum measurements from the Muon Spectrometer are combined with those from the Inner Detector to achieve improved resolution across a broad momentum range. The system provides robust muon identification and triggering capabilities, which are essential for many physics analyses.

1.3.5 Trigger and Data Acquisition System. The high collision rate delivered by the LHC far exceeds the rate at which events can be recorded for offline analysis. To address this challenge, ATLAS employs a trigger and data acquisition system that selects potentially interesting events in real time while rejecting the majority of collisions.

The trigger system reduces the event rate through a multi-level selection process based on information from the detector subsystems. Trigger decisions are designed to maintain high efficiency in the physics processes of interest while operating within strict latency and bandwidth constraints. The data acquisition system then records the selected events for further processing and offline reconstruction. Together, the trigger and data acquisition systems enable ATLAS to efficiently exploit the physics potential of the LHC.

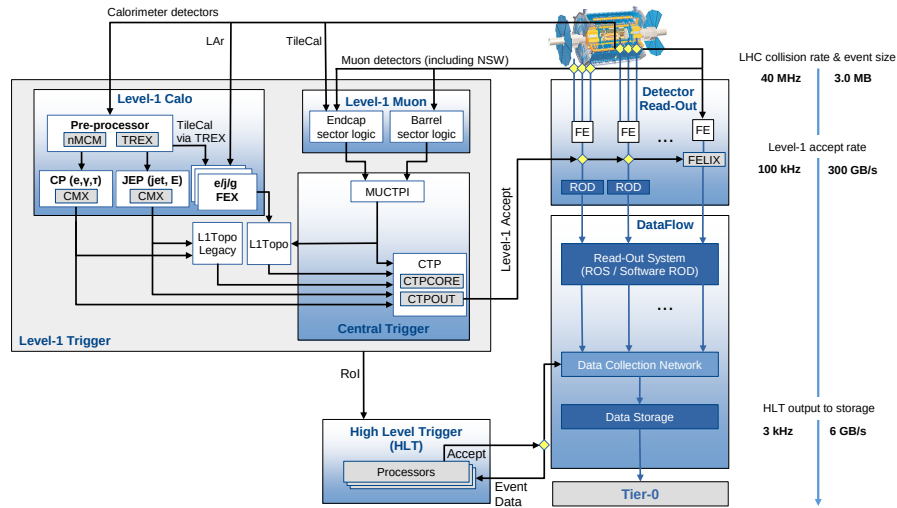


Figure 8. ATLAS Trigger and Data Acquisition System in Run 3 [22]

1.4 ATLAS Run 4 Upgrade

In Run 4, the Large Hadron Collider will enter the High-Luminosity LHC (HL-LHC) phase, which is designed to significantly extend the physics reach of the ATLAS experiment. The HL-LHC aims to deliver an integrated luminosity of approximately $3\text{--}4\text{ ab}^{-1}$ over its operational lifetime, corresponding to more than an order-of-magnitude increase relative to the datasets collected in earlier runs. This increase will enable precise measurements of rare Standard Model processes and enhance sensitivity to physics beyond the Standard Model.

The higher luminosity of the HL-LHC will result in substantially more challenging experimental conditions. The instantaneous luminosity increases to an average pile-up level of up to $\langle\mu\rangle \sim 200$ interactions per bunch crossing. Such conditions will significantly increase detector occupancies, radiation exposure, and the complexity of event reconstruction. Maintaining the performance of tracking, vertexing, and triggering under these conditions requires major upgrades to the ATLAS detector systems.

A key component of the Phase-II upgrade program is the complete replacement of the current Inner Detector with a new all-silicon Inner Tracker (ITk). The ITk is designed to provide increased granularity, improved radiation tolerance, and extended pseudorapidity coverage compared to the existing detector. These improvements are essential for preserving tracking efficiency, momentum resolution, and vertex reconstruction performance in the high-pile-up environment of the HL-LHC.

In parallel, the ATLAS trigger and data acquisition system will be upgraded to cope with the increased event rates and data volumes expected during HL-LHC operation. The upgraded trigger system is designed to support higher output

rates (10 kHz compared to 3 kHz in Run 3) and more sophisticated real-time reconstruction algorithms, enabling efficient selection of physics events while operating within strict latency constraints.

The ATLAS Phase-II upgrade program is therefore essential for sustaining detector performance and physics sensitivity throughout the HL-LHC era. The design choices and technological developments implemented during this upgrade phase directly address the challenges posed by high luminosity and pile-up [23] [24].

CHAPTER II

ATLAS EVENT FILTER (EF) TRACKING: STRIP CLUSTERING

2.1 Introduction

This report presents performance studies related to the Event Filter (EF) Tracking project of the ATLAS experiment, a part of its Phase-II upgrade program for the High-Luminosity LHC (HL-LHC). The HL-LHC pile-up conditions will produce an average of 200 inelastic proton–proton collisions per bunch crossing, and a new full-silicon Inner Tracker (ITk) [25, 26], designed to allow efficient tracking in such an environment, will be installed during the Phase-II upgrade. The high pile-up conditions represent a significant challenge to the ATLAS TDAQ system, whose architecture and physics motivations are described in the Phase-II TDAQ TDR [27] and the Phase-II TDAQ TDR Amendment [28]. All the EF algorithms, including the EF Tracking ones, will run on a flexible, heterogeneous commercial system comprising CPU cores, possibly assisted by commodity accelerators such as GPUs or FPGAs. The rate at which events can be reconstructed for a given amount of computational hardware is a critical design factor for the EF system, with tracking historically the most computationally intensive reconstruction performed at the EF stage. The TDAQ design foresees that events are first processed by a hardware trigger (Level-0), which bases its decision on calorimeter and muon detector data. Accepted events are passed to the EF, which gathers the data from all the detector readout systems at a maximum rate of 1 MHz. Following EF processing, the event acceptance rate is reduced to 10 kHz. Track reconstruction will play a critical role in the subsequent trigger decisions, helping to keep the EF output rate at a manageable level. It can be used to refine the selection of objects identified at Level-0, improve the precision of their kinematic measurements, and suppress

multi-object coincidences, for instance by requiring that associated tracks originate from the same primary vertex. Track reconstruction is, however, more challenging due to the high processing rate and the large hit density in the tracking detectors, which makes pattern recognition more computationally demanding. At the same time, the optimized layout of the ITk is expected to reduce the fake-track rate to a negligible level [25, 26].

To choose an EF Tracking design adapted to those challenges, the ATLAS Phase-II EF Tracking community embarked on a series of demonstrator cycles to study so-called tracking pipelines. In parallel, performance requirements for the EF tracking algorithms as well as the EF farm were produced [29, 30], motivated by PPES studies aimed at the HL-LHC physics goals of ATLAS [31]. Following this, the design selection criteria and process were established [32]. Both the requirements and criteria were reviewed and approved by a committee composed of various experts from within and outside of the ATLAS TDAQ community [33].

The studies shown in this report focus on exploring a possible EF Tracking design that makes use of GPU accelerators alongside conventional CPU processing. Tracking algorithms are put together to form a tracking pipeline. The pipeline referred to as C-000 is the default ITk track reconstruction which uses the legacy implementation of the tracking algorithms in ATHENA and serves as a reference for the comparisons of the various metrics. Several pipelines are described, with varying amounts of processing performed on GPU as indicated by the first digit: from G-0XX (minimal offloading) to G-2XX (full offloading).

2.1.1 Overview of GPU-based pipelines. The framework known as A Common Tracking Software (ACTS) [34] represents a robust, high-performance solution for track reconstruction in Particle Physics, usable across multiple

experiments. Designed with modularity, detector independence, and adaptability to contemporary computing architectures in mind, ACTS has been successfully integrated into the ATHENA software framework of the ATLAS experiment. It is currently slated to replace the existing offline tracking software for LHC Run 4 and subsequent operations. Furthermore, ACTS has been proposed as a key component for the Event Filter (EF) online track reconstruction, particularly within the C-100 and C-230 pipelines [35].

TRACCC [36] is a dedicated GPU-oriented research and development extension of ACTS, independently developed to address the stringent performance requirements of online track reconstruction in dense, high-rate detector environments. While TRACCC shares foundational algorithms, supporting libraries, and detector descriptions with ACTS, it is a separate piece of software designed to exploit massive parallelism inherent in GPU hardware and to achieve portability across varying GPU hardware and software frameworks. This report focuses on the deployment of TRACCC within multiple types of GPU-based processing pipelines.

Within the pipelines denoted G-2X0, the entire track reconstruction sequence—including clustering, space point formation, seeding, track finding, and track fitting—is executed on the GPU. The CPU is used solely for the transfer of reconstructed tracks from GPU memory to host memory and the conversion between TRACCC and ATHENA EDM. Other pipelines transition from GPU to CPU at different stages of the track reconstruction chain, executing the remaining reconstruction steps on the CPU. This GPU-CPU hybrid architecture leverages the strengths of both processing units to optimize overall performance, hypothesising that the most parallelizable and least branching stages—such as clustering and

seeding—benefit most from GPU acceleration. The remaining steps use the ACTS algorithms in a workflow equivalent to that of the C-100/C-230 pipelines. The hybrid pipelines differ in which stages of track reconstruction are offloaded to the GPU: G-0X0 pipelines offload only the clustering step, G-1X0 pipelines offload both clustering and track seeding, G-2X5 offload clustering, seeding and track finding. The modularity of the track reconstruction chain and TRACCC software allows for a step-by-step integration into ATHENA/ACTS and component-by-component validation while slowly increasing the usage of GPUs in the trigger. To showcase the performance of track reconstruction on GPUs we focus the main body of this report on three flagship pipelines: G-050, G-130 and G-200.

Track reconstruction is strongly affected by the quantity and quality of seeds produced in the corresponding step of the track reconstruction chain. While both TRACCC and ACTS provide a seed-finding algorithm, this report also presents an alternative algorithm known as Graph Based Track Seeding version 2 and we demonstrate its impact on the GPU based pipelines. The two different track seeding algorithms, the classical triplet seeder and graph-based seeding algorithm, are distinguished by the naming convention, namely adding 30 to the base pipeline label when the graph-based seeder is employed.

2.2 Overview of GPU-based pipelines

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Pipeline	EDM conv.	Clustering	Seeding	Track finding	Track fitting	EDM conv.
C-100/C-230	CPU	CPU	CPU	CPU	CPU	CPU
G-050/G-080	CPU	GPU	CPU	CPU	CPU	CPU
G-100/G-130	CPU	GPU	GPU	CPU	CPU	CPU
G-205/G-235	CPU	GPU	GPU	GPU	CPU	CPU
G-200/G-230	CPU	GPU	GPU	GPU	GPU	CPU
G-300	CPU	CPU	GPU (GNN)		CPU	CPU

Figure 9. Overview of the reconstruction pipelines, including the G-050, G-130, and G-200 pipelines discussed in this thesis.

2.2.0.1 Motivation for GPU Implementation.

Silicon strip clustering exhibits intrinsic parallelism. Clusters are formed independently within

each detector module, and individual strip hits can be processed concurrently when searching for neighboring channels above threshold. This structure makes the algorithm well-suited for GPU execution, where thousands of lightweight threads can operate simultaneously.

However, achieving efficient GPU performance requires more than simply parallelizing loops. Memory layout, coalesced access patterns, thread mapping strategies, and synchronization between neighboring strips must be carefully designed to avoid performance bottlenecks. Additionally, the algorithm must reproduce the clustering behavior of the reference Athena implementation to ensure consistency in downstream tracking performance.

The motivation for implementing strip clusterization on GPUs within `traccc` is therefore twofold:

- To reduce the computational latency of clustering in high-occupancy environments.
- To establish a validated GPU-based tracking workflow compatible with ATLAS software standards.

Demonstrating physics equivalence while achieving computational gains is essential for future integration into trigger or offline reconstruction chains.

2.2.1 Previous work and motivation.

2.2.2 Overview of GPU-based Pipelines.

2.2.3 Traccc and Athena framework.

2.2.4 Strip clustering algorithm. The clustering algorithm for the ITk silicon strip modules, as implemented in the `traccc` framework for the Event

Filter (EF) Tracking GPU pipelines, relies on classical component analysis. This approach is highly optimized for massive parallelism on GPGPU architectures.

The algorithm processes raw cell data from the detector and proceeds through the following steps:

1. **Component Analysis:** The algorithm scans for contiguous activated detector cells. A cluster is formed by grouping cells that are adjacent to one another in the sensor plane, characterized by their spatial indices.
2. **Weighted Centroid Calculation:** Once contiguous cells are identified, the cluster center is computed as a weighted centroid:

$$\vec{x}_{\text{local}} = \frac{\sum_i w_i \vec{x}_i}{\sum_i w_i} \quad (2.1)$$

where $\vec{x}_i$ represents the spatial indices of the i -th cell, and the weight w_i corresponds to the analog information of that cell. Unlike a pixel detector, a strip detector does not provide time-over-threshold information, so it produces a binary signal.

3. **Coordinate Transformation:** The local cluster measurement is transformed into physical coordinates on the detector module by referencing the cell pitch size and module dimensions, which are obtained from the digitization configuration file.
4. **Space Point Formation:** Local measurements are subsequently transformed into global space points using the transformation matrices and local coordinate systems specified in the pre-computed detector geometry description (handled via the `detray` library).

2.2.5 Input data.

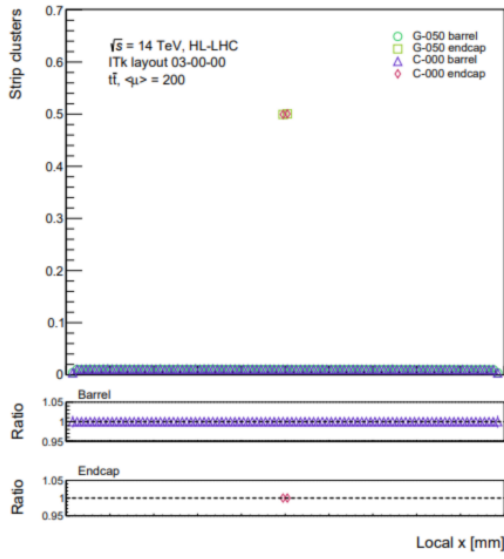
2.2.6 Integration with the Traccc framework.

2.3 Results and Discussion

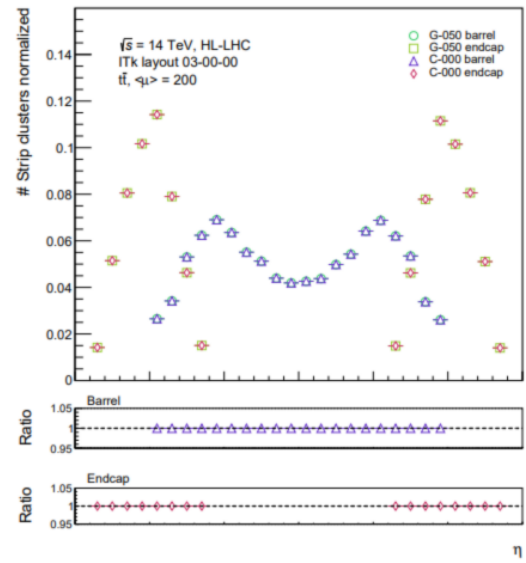
Generic Step	Step	$\mu = 140$		$\mu = 200$	
		Timing [ms]	[HS23.s]	Timing [ms]	[HS23.s]
RDO Conversion	RDO to traccc	121	4.2	159.0	5.6
GPU Transfer	Host to Device Transfer	1.1	0.0	1.2	0.0
Clustering	Clustering Algorithm	0.5		0.7	
Spacepoint Formation	Measurement Sorting	1.9		2.5	
	Cluster Conversion	35.9	1.3	47.1	1.7
Seeding	Seeding Algorithm	85.4	3.0	138.5	4.9
Track Finding	Track Finding Algorithm	236.5	8.3	336.7	11.8
Acts to xAOD Convers.	Conversion Algorithm	11.7	0.4	16.1	0.6
GPU Transfer	Device to Host Transfers	1.3	0.0	2.2	0.1
Total CPU		493.0	17.3	700.8	24.6
Of which: GPU overhead		159.3	5.6	209.5	7.4
Total GPU		2.4		3.1	
Accelerator Concurrency Factor		0.52		0.53	
Total GPU with Concurrency		1.3		1.6	

Table 1. Average timing of the different steps of track reconstruction using the G-O80 pipeline for $t\bar{t}$ events at $\langle\mu\rangle = 140$ and $\langle\mu\rangle = 200$.

2.3.1 Cluster comparison with CPU.

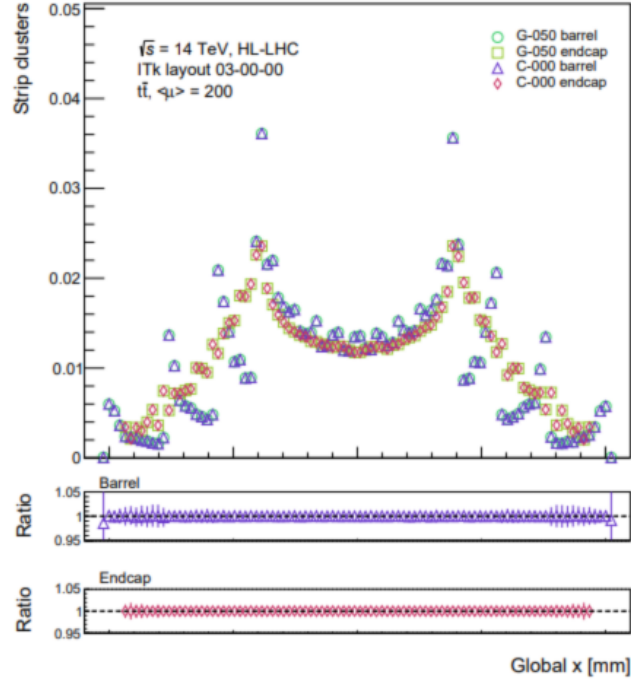


(a) Local position distribution.

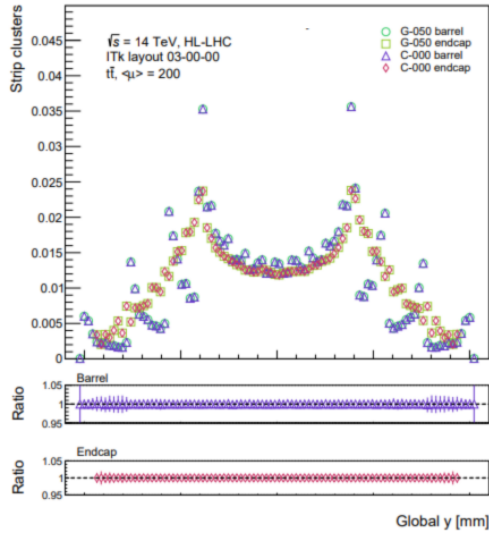


(b) Number of strip clusters vs. η .

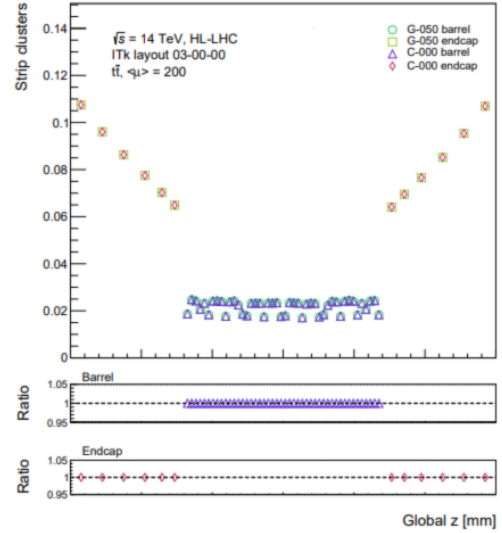
Figure 10. Validation of strip cluster local coordinates and η dependence.



(a) Global x position.



(b) Global y position.



(c) Global z position.

Figure 11. Global coordinate distributions for the strip clusters.

2.3.2 Efficiency.

2.3.3 Summary of EF Tracking validation.

CHAPTER III

ATLAS: SIGNAL OPTIMIZATION OF A SEARCH FOR HIGGS DECAYS TO DARK PHOTONS USING ATLAS RUN 3 DATA

3.1 Introduction

3.1.1 Motivation for Hidden Sectors. While Section 1.2 gives a brief overview of the astrophysical evidence for dark matter, the Standard Model offers no viable particle candidate. This motivates the hypothesis of a Dark Sector (DS), in which the dark photon (γ_d) is the gauge boson of an additional $U(1)_D$ gauge group. Strong motivations for the search of this particle arise from its potential contribution in solving the small-scale structure formation problems [38] [39], as well as being a possible source for the Sommerfeld enhancement of DM annihilation required to explain the PAMELA-Fermi-MS2 positron anomaly in terms of DM [40], or making asymmetric DM scenarios viable [41].

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3.1.2 Kinetic Mixing and the Dark Photon. The dark photon, denoted A'_μ , is associated with a new $U(1)_D$ gauge symmetry. The most general renormalizable Lagrangian describing the interaction between the SM hypercharge gauge field B_μ and the dark photon includes a kinetic mixing term [42]:

$$\mathcal{L} \supset -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{4}F'_{\mu\nu}F'^{\mu\nu} - \frac{\epsilon}{2}F_{\mu\nu}F'^{\mu\nu}, \quad (3.1)$$

where $F_{\mu\nu}$ and $F'_{\mu\nu}$ are the field strength tensors of the SM electromagnetic field and the dark photon field, respectively, and ϵ is the kinetic mixing parameter.

This interaction is gauge invariant and can be generated radiatively through loops of heavy particles charged under both $U(1)_Y$ and $U(1)_D$. Natural values of ϵ span a wide range, typically $10^{-12} \lesssim \epsilon \lesssim 10^{-2}$, depending on the ultraviolet completion of the theory [43].

After diagonalizing the kinetic terms and performing electroweak symmetry breaking, the dark photon acquires an effective coupling to the electromagnetic current:

$$\mathcal{L}_{\text{int}} = \epsilon e A'_\mu J_{\text{EM}}^\mu, \quad (3.2)$$

allowing the dark photon to interact weakly with electrically charged SM particles.

—

The decay of the Higgs boson into a photon and a dark photon, $H \rightarrow \gamma\gamma_d$, provides an interesting production channel for this BSM particle. While γ_d searches typically target the massive γ_d [?, ?, 44], which can couple directly to the Standard Model (SM) via kinetic mixing with the ordinary photon [45], this paper focuses on the massless γ_d scenario. In this case, the kinetic mixing between the γ_d the photon can be tuned away through a rotation of the gauge fields, and the γ_d can interact with ordinary matter only through operators of dimension higher than four [45]. The Higgs boson can decay through a loop interaction involving additional (pseudo-)scalar BSM messenger fields, which couple both to the SM and the DS [46, 47], as shown in Fig. 12. Thanks to the non-decoupling properties of the Higgs boson, and depending on the parameters of the model, a branching ratio (BR) of $H \rightarrow \gamma\gamma_d$ of few percent can be allowed [48, 49].

Searches for the $\gamma + E_{\text{T}}^{\text{miss}} + X$ signature have been widely performed at the LHC by the ATLAS and CMS experiments, in both the vector boson fusion (VBF) production mode and associated production with a Z boson (ZH) [?, ?, ?, ?, 50–52]. A combination of ATLAS results in the ZH and VBF channels currently provides the most stringent limits, excluding $\text{BR}(H \rightarrow \gamma\gamma_d) > 1.3\%$ (1.5% expected) at 95% CL. This result is based on an integrated luminosity of 139 fb^{-1} collected during LHC Run 2 at $\sqrt{s} = 13 \text{ TeV}$ [53].

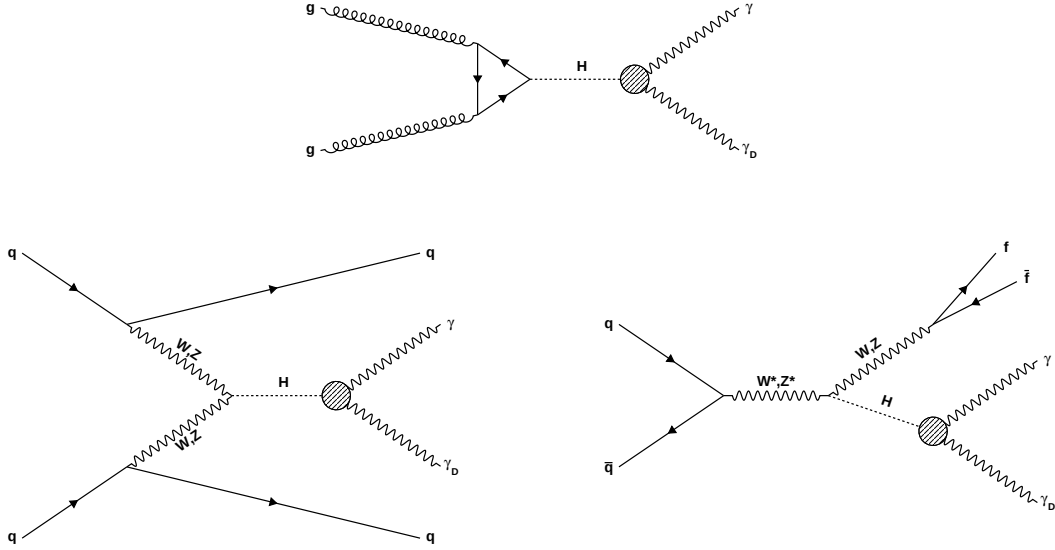


Figure 12. Feynman diagram illustrating the Higgs boson production via gluon-gluon fusion process (top) and vector-boson fusion (VBF) and Higgsstrahlung (VH) processes (bottom left and right), all followed by the decay into a photon and a dark photon ($H \rightarrow \gamma\gamma_d$). The hatched circle in the $H\gamma\gamma_d$ vertex represents the interaction through messenger fields.

Though these two production modes provide clean final states, the gluon-gluon fusion one (ggF) is an order of magnitude larger in cross-section than the VBF channel. This channel has been explored by CMS [?], but not yet by ATLAS – though the inclusive signature with a single photon has been pursued in the context of other exotic models [51]. One challenge is that this kind of signature was not accessible via the available triggers due to their high photon p_T or E_T^{miss} thresholds. Developments in the trigger between Run-2 and Run-3 led to the implementation of a lower threshold trigger, which increases the acceptance of ggF events by approximately a factor of two, while keeping the trigger rate under control.

The analysis presented in this paper targets the $H \rightarrow \gamma\gamma_d$ decay in the ggF production channel for the first time in ATLAS, retaining sensitivity also to the VBF, ZH and WH channels. The analysis is based on the LHC Run-3 proton–proton collisions data at $\sqrt{s} = 13.6$ TeV recorded by the ATLAS experiment in 2023 and 2024, corresponding to an integrated luminosity of 134.6 fb^{-1} .

3.2 Data and MC samples

3.2.1 Data samples. The data used in this analysis were collected in 2023 and 2024. The used Good Run List is

Year	GRL String	Lumi [fb^{-1}]
2023	data23_13p6TeV.periodAllYear_DetStatus-v133-pro31-11_MERGED_PHYS.StandardGRL.All.Good.25ns	25.2
2024	data24_13p6TeV.periodsEto0_DetStatus-v130-pro36-08_MERGED_PHYS.StandardGRL.All.Good.25ns	109.4

Table 2. Summary of the Good Run Lists (GRL) and integrated luminosities used for the Run 3 analysis.

for a total integrated luminosity of 134.6 fb^{-1} , with an uncertainty of 2% [54].

3.2.2 MC samples. The MC samples were processed with p-tag p6697.

The signal samples were showered from common Higgs LHE samples generated using Powheg. The decay process used in the shower was implemented with Pythia. The relevant JIRA tickets can be found at: ATLMCPROD-10733, ATLMCPROD-10819, ATLMCPROD-10916 and the sample validation at: MC23HLRSHyydValidation. The signal sample paths are shown in table 2.

The primary backgrounds for the analysis include processes with a true photon in the final states and genuine E_T^{miss} . These samples include: $Z\gamma$ and $W\gamma$, where the vector boson can decay to a lepton that is not reconstructed or to neutrinos. Additional background processes arise from fake photons due to misidentified jets or electrons, and/or fake E_T^{miss} . The background due to jets or electrons misidentified as photons is mainly given by $Zjets$, $Wjets$, di-jets and $\gamma jets$ with fragmentation photon. The $Zjets$, di-jets and $\gamma(frag)jets$ events are also characterized by fake E_T^{miss} due to mis-measured or not reconstructed jets. Finally, $\gamma jets$ with direct photon ($\gamma(direct)jets$) enter the analysis regions exclusively due to the presence of fake E_T^{miss} .

The list of MC samples used in this analysis is presented in the Appendix tables. As detailed later, MC simulations are eventually employed only for true photon backgrounds (including the $\gamma(direct)jets$), while the contribution from events with fake photons is estimated through data-driven methods. The W/Z MC samples are used as part of the jet-fakemethod and therefore listed in the following tables, while $\gamma(direct)jets$ and di-jets have been used exclusively for preliminary studies and initial optimization of the analysis selection and are not reported here.

3.3 Physics object reconstruction and identification

Physics objects used in this analysis are reconstructed following standard ATLAS procedures. Photon candidates are reconstructed from topological clusters in the electromagnetic calorimeter and calibrated using an energy calibration chain, and are required to satisfy tight identification criteria based on shower-shape variables together with isolation requirements designed to suppress hadronic backgrounds [55]. Muons are reconstructed by combining tracks from the inner detector and the muon spectrometer, and are selected using quality and isolation working points that ensure high efficiency while suppressing non-prompt backgrounds [56]. Electrons are formed by matching inner-detector tracks to electromagnetic calorimeter clusters and are identified using likelihood-based discriminants exploiting shower shapes, track quality, and track-cluster matching observables [57]. Jets are reconstructed using the anti- k_t algorithm with radius parameter $R = 0.4$ applied to particle-flow or topological calorimeter inputs, and are calibrated through simulation-based and in situ corrections, with additional pile-up mitigation techniques such as jet-vertex tagging applied [58]. To avoid double-counting of detector signatures, reconstructed objects are subjected to an overlap-removal procedure that resolves ambiguities among nearby photons, leptons, and jets using a fixed priority scheme [59]. The missing transverse momentum, $\vec{p}_T^{\text{miss}}$, is computed as the negative vector sum of the transverse momenta of all reconstructed baseline objects, supplemented by a track-based soft term associated with the primary vertex to reduce pile-up sensitivity [60]. These definitions ensure a consistent and well-validated reconstruction of final-state objects across ATLAS analyses.

Table 3. Electron selection criteria.

Feature	Criterion
Object quality	Not from a bad calorimeter cluster (<code>BADCLUSELECTRON</code>)
Energy calibration	<code>es2024_Run3_v0</code> (ESModel)
Pseudorapidity range	$(\eta < 1.37) \quad \quad (1.52 < \eta < 2.47)$
Transverse momentum	$p_T > 10 \text{ GeV}$
Track to vertex association	$ d_0^{\text{BL}}(\sigma) < 5$ $ \Delta z_0^{\text{BL}} \sin \theta < 0.5 \text{ mm}$
Baseline	
Identification	<code>LooseAndBLayerLLH</code>
Selected	
Identification	<code>TightLLH</code>
Isolation	<code>Tight_VarRad</code>

Table 4. Jet reconstruction criteria.

Feature	Criterion
Algorithm	<code>Anti--kt</code>
R -parameter	0.4
Input constituent	PFlow
Calibration configuration	<code>AntiKt4EMPFLOW_MC23a_PreRecR22_Phase2_CalibConfig_ResPU_EtaJES_GSC_241208_InSitu.config</code>
Jet cleaning	<code>LooseBad</code>
p_T	$p_T > 20 \text{ GeV}$
$ \eta $	< 4.5
JVT Working point	<code>FixedEffPt</code>
fJVT Working point	<code>Loose</code>

Table 5. Muon selection criteria.

Feature	Criterion
Object quality	Bad Muon Veto
p_T cut	$p_T > 10$ GeV
$ \eta $ cut	< 2.7
d_0 significance cut	3
z_0 cut	0.5 mm
Baseline	
Identification working point	Medium
Selected	
Identification working point	Medium
Isolation working point	PflowLoose_VarRad

Table 6. Overlap removal criteria.

Reject	Against	Criteria
Electron	Electron	shared track, $p_{T,1} < p_{T,2}$
Tau	Electron	$\Delta R < 0.2$
Tau	Muon	$\Delta R < 0.2$
Muon	Electron	is Calo-Muon and shared ID track
Electron	Muon	shared ID track
Photon	Electron	$\Delta R < 0.4$
Photon	Muon	$\Delta R < 0.4$
Jet	Electron	$\Delta R < 0.2$
Electron	Jet	$0.2 < \Delta R < 0.4$
Jet	Muon	NumTrack < 3 and (ghost-associated or $\Delta R < 0.2$)
Muon	Jet	None
Jet	Tau	$\Delta R < 0.2$
Jet	Photon	$\Delta R < 0.4$

Table 7. Photon selection criteria.

Feature	Criterion
Object quality	Not from a bad calorimeter cluster (<code>BADCLUSPHOTON</code>)
Photon cleaning	<code>pass0Qquality</code>
Pseudorapidity range	$(\eta < 1.37) \quad \quad (1.52 < \eta < 2.37)$
Energy calibration	<code>es2024.Run3.v0</code> (ESModel)
Baseline photons	
Transverse energy	$p_T > 25 \text{ GeV}$
Identification	Tight
Selected photon	
Transverse energy	$p_T > 50 \text{ GeV}$
Identification	Tight
Isolation	TightCaloOnly

3.4 Event selection

3.4.1 Trigger. The analysis uses a trigger implemented in 2023, see ATR-26410. The signal trigger requires a single photon with 26 GeV at L1 – which is shared with the single electron L1 item. The HLT is a multi-object trigger that requires a 50 GeV tight ID photon (g50), 40 GeV of cell E_T^{miss} (xe40_cell) and 70 GeV of “pfopufit” E_T^{miss} (xe70_pfopufit). There transverse mass computed between the photon and the E_T^{miss} is defined using the expression: $m_T = \sqrt{2E_T^\gamma E_T^{miss}(1 - \cos(\Delta\phi))}$. The trigger requires that: $m_T > 80$ GeV be applied (80mTAC).

This trigger is consistently applied in all the analysis regions, which are generally characterized by signatures allowing reasonable trigger acceptance: $1\mu + \gamma$, $2\mu + \gamma$, 1 non-isolated $\gamma + E_T^{miss}$, $1e + E_T^{miss}$. For the muon-CRs, it is possible to apply the analysis trigger taking advantage of the fact that muons are not included in the trigger-level E_T^{miss} computation. The most critical final state is $1e + E_T^{miss}$: in this case, an OR of multiple single-electron triggers and the analysis trigger is applied.

Table 8. Triggers used in the analysis.

Use	Trigger name
All regions	HLT_g50_tight_xe40_cell_xe70_pfopufit_80mTAC_L1eEM26M
$1e + E_T^{miss}$ region	OR of single-e triggers and HLT_g50_tight_xe40_cell_xe70_pfopufit_80mTAC_L1eEM26M

3.5 Signal region optimization

The set of rectangular cuts resulting from the previous optimization is the following:

Preselections:

- Analysis trigger: HLT_50ightz40_eLLz70_fopupfit8_mTAC_LeEM26M
- $N_\gamma = 1$, with N_γ the number of selected photons;

- $N_{\text{leptons}} = 0$, for baseline leptons;
- $N_{\text{jet, central}} \leq 3$;
- $p_T^\gamma > 50 \text{ GeV}$;
- $E_T^{\text{miss}} > 100 \text{ GeV}$;
- $m_T(E_T^{\text{miss}}, \gamma) > 80 \text{ GeV}$;

Additional selections:

- $\text{BDT}_{\text{score}}^{\text{vertex}} > 0.1$;
- E_T^{miss} significance > 6 ;
- $\Delta E_T^{\text{miss}} > -10 \text{ GeV}$;
- $\Delta\phi(E_T^{\text{miss}}, E_T^\gamma) > 1.25$;
- $\Delta\phi(E_T^{\text{miss}}, E_T^{\text{jet}}) < 0.75$;
- $\Delta\phi(j_1, j_2) < 2.5$;
- $|\eta_\gamma| < 1.75$;

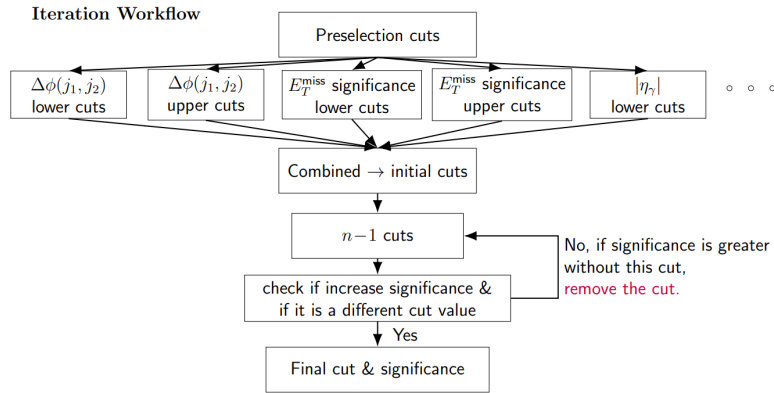


Figure 13. Iteration workflow for optimizing the cut set using n-1 studies and significance checks.

3.5.1 N-1 iteration method. The iteration process, demonstrated in figure 13, determines the optimized cuts for the additional selections. After

applying preselection cuts, it imposes lower and upper bounds on the key variables, denoted c_i for clarity, based on their ROC curves. If either an upper/lower cut does not improve significance, we will skip the cut for this variable, but it will be reconsidered and iterated in a later process. Combining all the cuts becomes our initial cuts of the variables. Then it performs n-1 cuts: for each cut c_i , apply all the cuts except c_i and calculate the n-1 significance. Choose the cut value of c_i that maximizes the significance, and repeat the process for the next c_{i+1} . If the n-1 significance is greater without the cut, we remove the cut. In the next iteration, it will reconsider the entire list of cut variables and cut values, including those previously dropped. We finalize the optimized cuts and significance if none of the chosen cuts change, and there is no further improvement in significance in one iteration.

After checking n-1 plots of each variable, we optimized the cuts to:

The balance variable is defined as

$$\text{balance} \equiv \frac{E_T^{\text{miss}} + p_T^{\text{leading } \gamma}}{\sum_{\text{central jets}} p_T^j}. \quad (3.3)$$

Compared to the internal note cuts, the lower cut of E_T^{miss} significance change from 6 to 7, the ΔE_T^{miss} alters from -10 GeV to -25 GeV, and we add a lower cut of 0.90 for balance.

With the improved optimized cuts, we increased the simple significance $\frac{S}{\sqrt{B}}$ from 2.475 to 2.701.

Table 9. Preselections and basic-cut comparison between the Internal Note and Optimized Selections

Selection	Internal Note	Optimized Selections
<i>Preselections</i>		
Analysis trigger	HLT_50ightz40_eLLz70_fopupfit8_mTAC_LeEM26M	
Number of photons N_γ		$= 1$
Number of leptons N_{leptons}		$= 0$ (baseline leptons)
Central jets $N_{\text{jet, central}}$		≤ 3
Photon transverse momentum p_T^γ		> 50 GeV
Missing transverse energy E_T^{miss}		> 100 GeV
Transverse mass $m_T(E_T^{\text{miss}}, \gamma)$		$100 < m_T < 140$ GeV
<i>Additional selections</i>		
BDT _{score} ^{vertex}	> 0.1	> 0.1
E_T^{miss} significance	> 6	> 7
Photon pseudorapidity $ \eta_\gamma $	< 1.75	< 1.75
$\Delta\phi(E_T^{\text{miss}}, E_T^\gamma)$	> 1.25	> 1.25
ΔE_T^{miss}	> -10 GeV	> -25 GeV
$\Delta\phi(j_1, j_2)$ (central jets)	< 2.5	< 2.5
$\Delta\phi(E_T^{\text{miss}}, \text{jet})$	< 0.75	< 0.75
balance	—	> 0.90
$S/\sqrt{B}$	2.475	2.701

3.5.2 Statistical method. The combination presented in this section relies on a combined likelihood function defined as the product of the likelihoods from the individual channels entering the combination, which are themselves products of likelihoods computed from the final observables in various categories in a single analysis.

The common parameter of interest (POI) targeted in this combination is the branching ratio of $H \rightarrow \gamma\gamma_d$, denoted by μ . Systematic uncertainties are included as nuisance parameters, denoted by θ , and are constrained by Gaussian probability density functions. These encode information from *auxiliary* measurements and measure the effect of systematic uncertainties. The likelihood also includes normalization factors, denoted by κ , which are floated in the fit

without constraints to adjust the agreement with data of background components in their corresponding CR(s).

The combined likelihood can be written as [61]

$$\mathcal{L}(\text{data}|\mu, \boldsymbol{\kappa}, \boldsymbol{\theta}) = \prod_{c=1}^{N_{\text{cats}}} \mathcal{L}_c(\text{data}|\mu, \boldsymbol{\kappa}, \boldsymbol{\theta}) \prod_{k=1}^{N_{\text{cons}}} \mathcal{G}(\tilde{\theta}_k|\theta_k), \quad (3.4)$$

where N_{cats} is the number of categories, N_{cons} is the number of constrained NPs, $\tilde{\theta}_k$ is the global observable corresponding to θ_k , c is the index for the event categories, k is the index for the constrained NPs, and $\mathcal{G}$ represents a Gaussian distribution.

The likelihood for a given category $\mathcal{L}_c$ can be expressed as the product of Poisson distributions, one for each bin in the histogram of the observable considered for that category, as

$$\mathcal{L}_c(\text{data}|\mu, \boldsymbol{\kappa}, \boldsymbol{\theta}) = \prod_{i=1}^{N_c^{\text{bins}}} \mathcal{P}\left(n_c^i \left| \sum_S S_c^i(\mu, \boldsymbol{\theta}) + \sum_B B_c^i(\boldsymbol{\kappa}, \boldsymbol{\theta}) \right.\right), \quad (3.5)$$

where N_c^{bins} is the number of bins, i is the bin index, $\mathcal{P}$ denotes a Poisson distribution, n_c^i is the number of (observed or Asimov) data events in each bin, S_c^i and B_c^i are the expected yields of a given signal sample S and background sample B , respectively, in each bin.

The CL_s frequentist method [62] following the asymptotic formulae [63] is used to derive the 95% CL upper limit on the POI with the profile likelihood ratio test statistic (q_μ), defined as

$$q_\mu = -2 \ln \frac{\mathcal{L}(\mu, \hat{\boldsymbol{\kappa}}(\mu), \hat{\boldsymbol{\theta}}(\mu))}{\mathcal{L}(\hat{\mu}, \hat{\boldsymbol{\kappa}}, \hat{\boldsymbol{\theta}})}, \quad (3.6)$$

where the numerator indicates the values of $\boldsymbol{\kappa}$ and $\boldsymbol{\theta}$ that maximize $\mathcal{L}$ for a given value of μ , and the denominator is evaluated for the values $\hat{\mu}$, $\hat{\boldsymbol{\kappa}}$, $\hat{\boldsymbol{\theta}}$ which jointly maximize the likelihood.

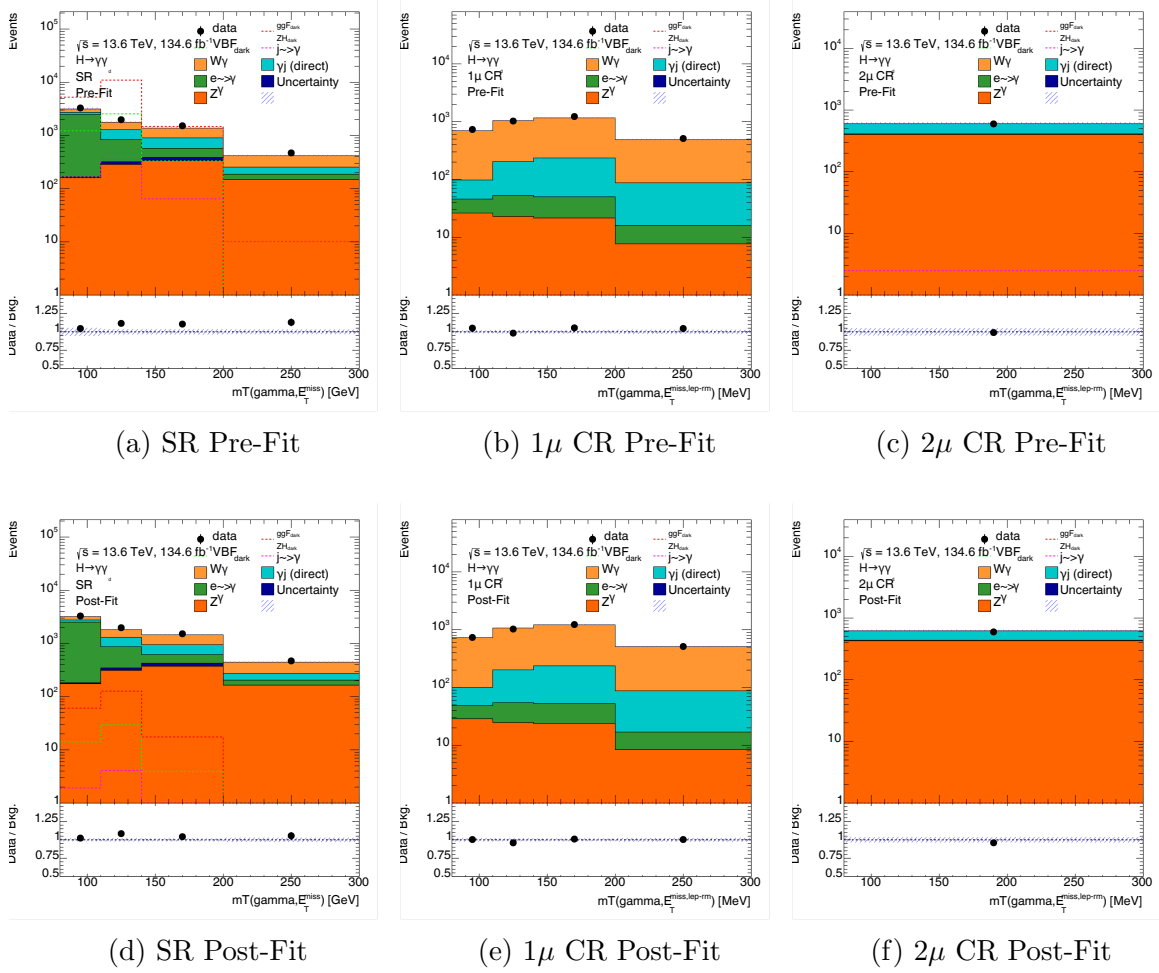


Figure 14. Generated by Tim Matthew (senior PhD at University of Oregon), pre-fit (top row) and post-fit (bottom row) m_T distributions in the Signal Region (SR), 1μ Control Region, and 2μ Control Region. The distributions span 80–300 GeV with binning (80, 110, 140, 200, 300) GeV.

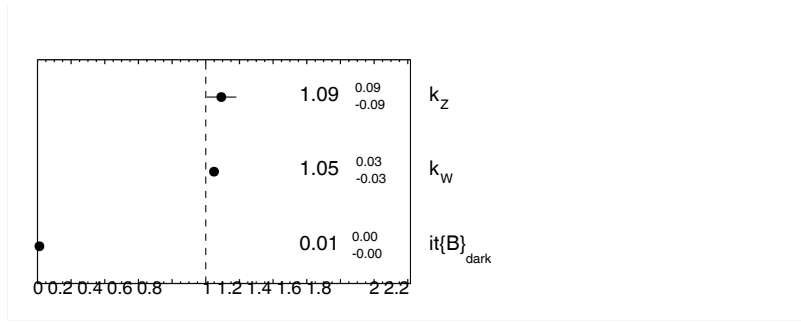


Figure 15. Normalization factors for W and Z processes and signal strength, with related statistical uncertainties

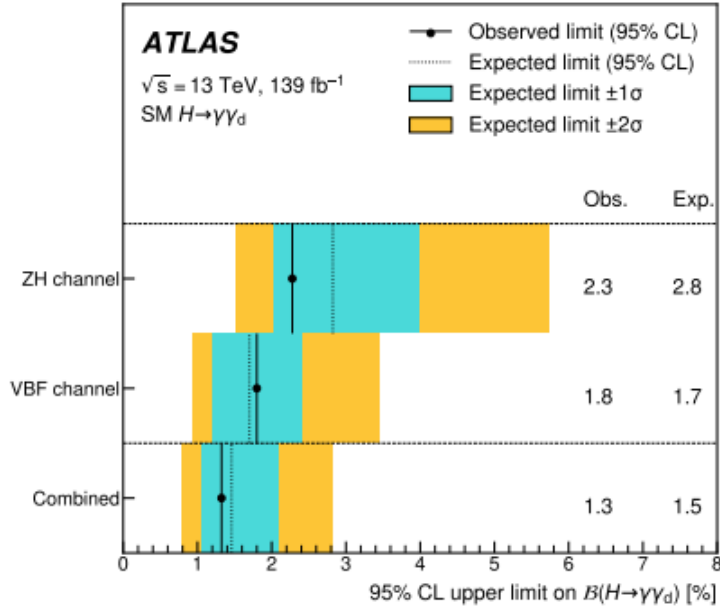


Figure 16. Observed and expected 95% CL upper limits on the branching ratio $\mathcal{B}(H_{125} \rightarrow \gamma\gamma_D)$ obtained from the individual analysis channels and their combination. The black markers indicate the observed limits, while the dashed line represents the median expected limit under the background-only hypothesis. The green and yellow bands correspond to the $\pm 1\sigma$ and $\pm 2\sigma$ intervals on the expected limit, respectively [64].

A comparison of the m_T distributions before and after the application of additional selection cuts is shown in Figure 14. In this context, the “pre” distributions correspond to the baseline event selection, while the “post” distributions are obtained after applying the additional optimized requirements.

The comparison is performed in the Signal Region (SR), the 1μ Control Region (CR), and the 2μ Control Region. The m_T distributions span 80–300 GeV with binning (80, 110, 140, 200, 300) GeV. The additional selection requirements are designed to enhance signal sensitivity in the SR while maintaining control over the dominant background contributions in the CRs.

As expected, the impact of the additional cuts is most visible in the SR, where a reduction of background yields is observed at higher m_T , improving the signal-to-background separation. In the control regions, the overall normalization and shapes of the dominant backgrounds remain stable, indicating that the additional requirements do not introduce significant distortions to the background modeling. In the 2μ CR, where an inclusive binning scheme is used due to limited statistics, the effect of the additional cuts is primarily reflected in the overall yield.

The fitted normalization factors k_W and k_Z , together with their associated statistical uncertainties, are shown in Figure 15. Both normalization factors are close to unity, indicating good agreement between the predicted and observed yields in the control regions. The uncertainties are at the level of $\mathcal{O}(10\%)$, reflecting the statistical precision of the control samples. No significant deviation from the nominal expectation is observed, and the fitted values remain compatible with one another within uncertainties. This confirms that the background modeling for the $W\gamma$ and $Z\gamma$ processes is well constrained by the data.

Overall, the post-selection distributions show that the optimized cuts improve discrimination in the SR while preserving the integrity of the control regions.

The most recent ATLAS combination [64] reports an observed (expected) 95% confidence level (CL) upper limit on the branching ratio $\mathcal{B}(H_{125} \rightarrow \gamma\gamma_D)$ of 1.3% (1.5%), shown in Figure 16. Using the optimized selection strategy described in this work, the observed 95% CL upper limit is reduced to 0.88%, corresponding to an improvement of approximately 32% in the excluded branching ratio. This reduction reflects the enhanced background rejection achieved by the additional selection requirements while maintaining signal efficiency.

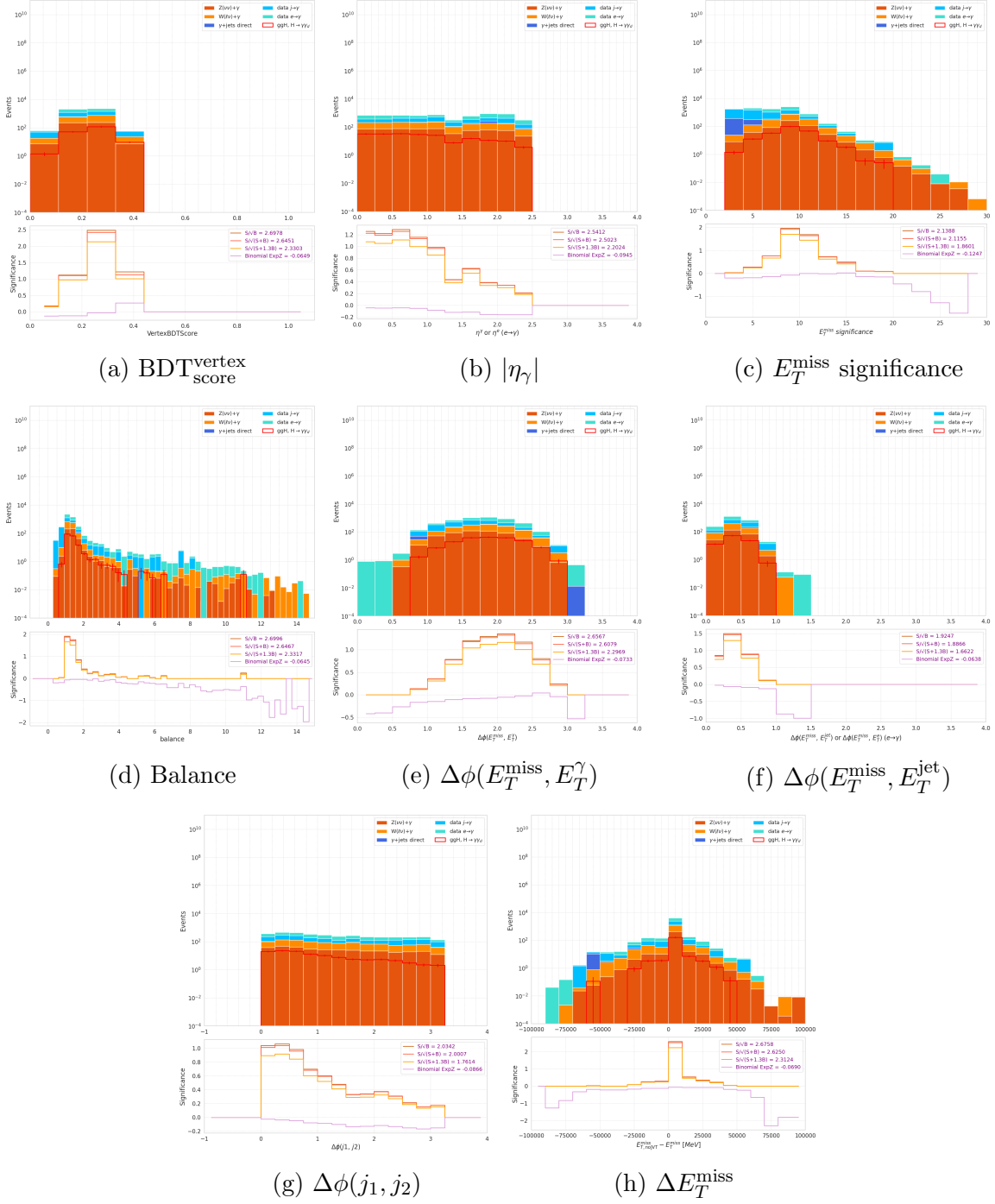


Figure 17. Additional Selection variables distributions after the optimized rectangular cuts for the ggF $H \rightarrow \gamma\gamma_D$ analysis.

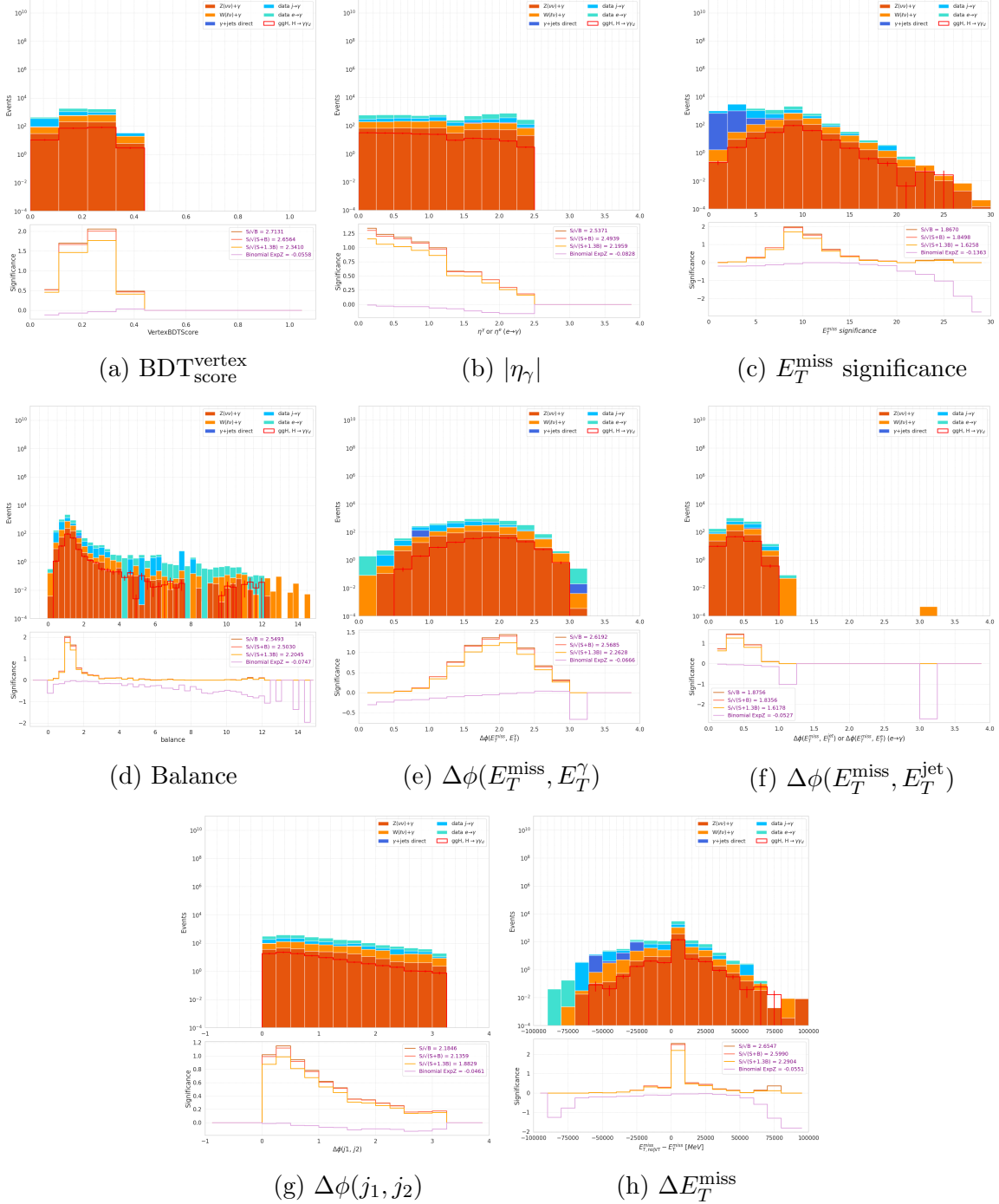


Figure 18. N-1 distributions for the ggF $H \rightarrow \gamma\gamma_D$ analysis.

3.5.3 Machine Learning approach.

3.5.3.1 Boosted Decision Tree (BDT) training and results.

To improve the separation of signal and background beyond cut-based selections, a multivariate boosted decision tree (BDT) classifier was trained using gradient boosting [65–67]. Training is performed after preselections and object definitions. The input variables were selected based on Receiver Operating Characteristic (ROC) curves, which indicated their discriminating power for cuts. If the area under the curve (AUC) of the ROC curve is close to 0.5, then the variable is excluded from the list for ML input. To ensure the ML is not biased towards the wrong vertices, $\text{BDT}_{\text{score}}^{\text{vertex}} > 0.1$ is added in the preselections. The input variables used in the BDT training are $\text{BDT}_{\text{score}}^{\text{vertex}}$, ΔE_T^{miss} , $\Delta\phi(j_1, j_2)$, $\Delta\phi(E_T^{\text{miss}}, j_{\text{central}})$, $\Delta\phi(E_T^{\text{miss}}, \gamma)$, $p_{T,j,\text{central}}^{(2)}$, E_T^{miss} significance, balance, η_γ , and p_T^γ .

Input validation and correlations. To ensure that the training is not dominated by redundant information, correlations among the input variables are inspected using the Pearson correlation matrix (Fig. 19). Moderate correlations are expected among E_T^{miss} -related observables.

Training strategy and hyperparameter scan. The BDT is trained with gradient boosting, and a randomized hyperparameter scan is used to assess stability and optimize performance [65, 67]. The scan explores variations in tree depth, learning rate, number of estimators, subsampling, and regularization. I performed a randomized sweep of 100 BDT runs to balance performance and training stability. The parameter distribution for max depth ranged from 3 to 10; the learning rate was drawn from a uniform distribution between 0.001 and 0.1; the number of estimators was a randomized integer between 100 and 900; the subsample was

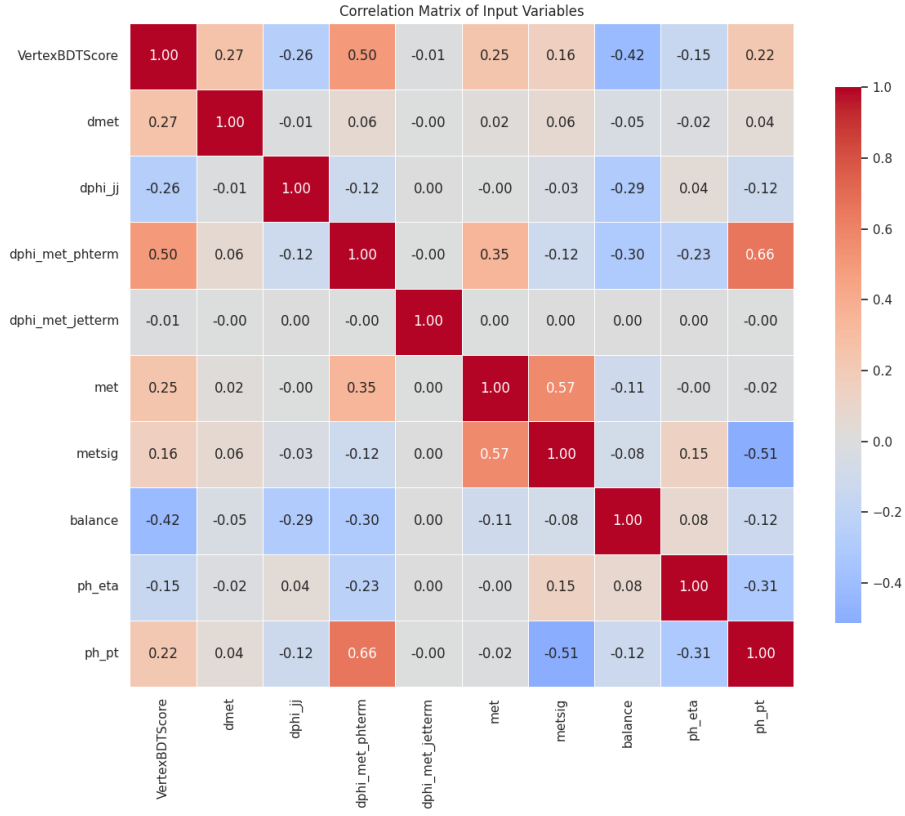


Figure 19. Correlation matrix of the BDT input variables after the baseline selection. The absolute value of the correlation above 0.75 will drop one of the two variables based on their physical significance and their AUC of ROC.

Table 10. Hyperparameter distribution in the randomized BDT sweep and the parameters for the best run

Hyperparameter	Scan Range / Distribution	Best Run
Maximum depth	randint(3, 10)	7
Learning rate	uniform(0.001, 0.1)	0.02399
Number of estimators	randint(100, 900)	398
Subsample fraction	uniform(0.2, 0.8)	0.7805
γ (split regularization)	uniform(0.1, 1)	0.2707
Best ROC AUC	0.957	
Signal Acceptance	48.63%	
BDT Score cut	0.02	
Maximum $S/\sqrt{B}$	2.960	

sampled from a uniform distribution between 0.8 and 0.2; the gamma was drawn from between 0.1 and 1.

Besides validation accuracy, I also looped over the cuts on the output ML BDT score and calculated the maximum simple significance ($S/\sqrt{B}$). The full sweep is in the Figure 20, where every run is recorded for its maximum simple significance and its corresponding ML BDT score cut and acceptance rate. The best-performing run (also in Table 10) is the one with the highest simple significance, using an ML BDT score cut ≥ 0.02 , which yields an expected significance of $S/\sqrt{B} = 2.960$ with a signal acceptance of approximately 48.63% in Fig. 21. The reason for not selecting ML BDT score cut ≥ 0.03 , despite their higher significance, is to avoid over-training and the loss of statistical sensitivity when performing the TRExFitter later.

The discriminating power of the BDT classifier is demonstrated by the ROC curve and the input feature importance to the classifier shown in Figure 22.



Figure 20. Weight and Biases plot [68] for the 95 BDT Sweeps (null runs were excluded), where it records the hyperparameters (gamma, learning rate, max depth, number of estimators, subsample) and performance (test AUC of the ROC, best $\frac{S}{\sqrt{B}}$, its corresponding ML BDT Score cut and acceptance rate). The highlighted line is the chosen parameter in table 10.

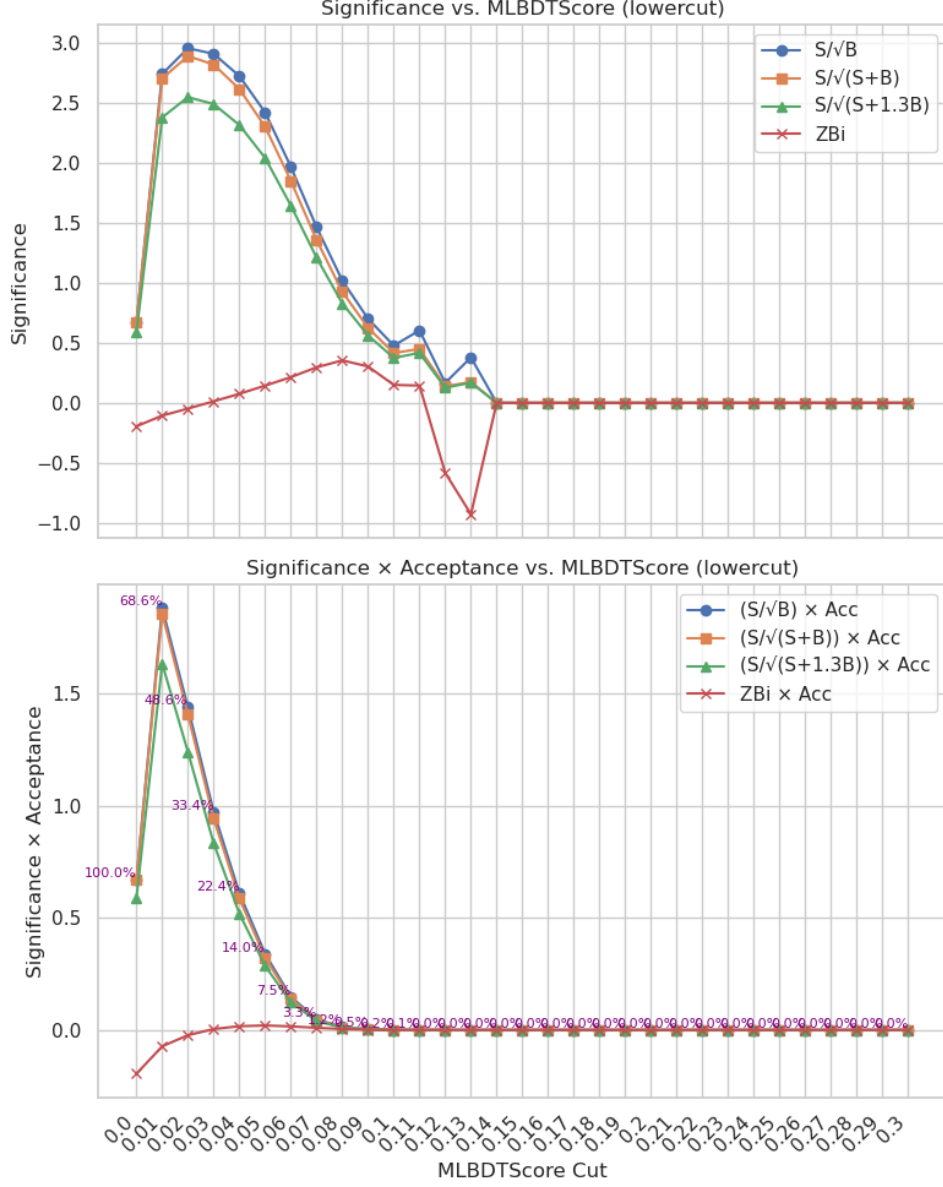
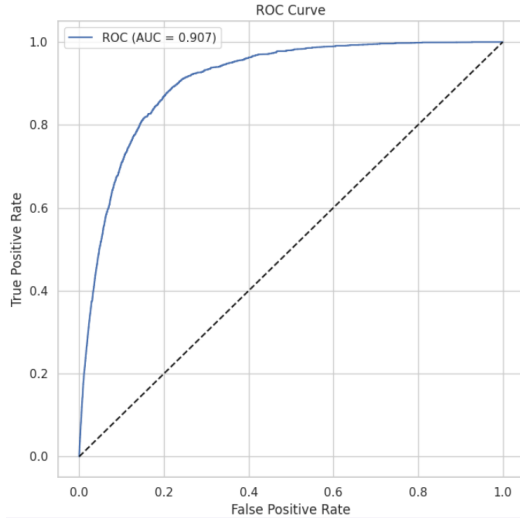
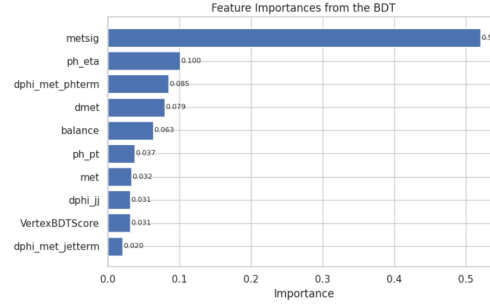


Figure 21. Significance versus BDT score threshold, with the maximum simple significance at BDT score ≥ 0.02 . $S/\sqrt{B}$ (blue) is the simple counting significance; $S/\sqrt{S+B}$ is (orange) the poisson-aware approximation; $S/\sqrt{S+1.3B}$ (green) is the Ad-hoc systematic-aware metric, assuming 30% background systematic; Zbi (red) is the Binomial significance.



(a) ROC curve of the trained BDT.



(b) Feature importance ranking from the trained model.

Figure 22. Performance of the BDT classifier. (Left) Receiver operating characteristic (ROC) curve. (Right) Feature importance of input variables.

3.5.3.2 *TRExFitter for BDT.*

3.5.3.3 Neural Network (NN) training and results. In addition to the BDT study, a 5-layer fully connected Deep Neural Network (DNN) classifier was trained to compare performance. DNN uses the same set of input variables as BDT, and then standardizes the input features, since neural networks are more sensitive to biases than BDT. The model is presented in the table 11. To optimize the model with the chosen hyperparameters (number of neurons per layer, learning rate, batch size, and dropout rate), a Bayesian hyperparameter sweep with 50 runs was performed and summarized in Table 12. Unlike the randomized sweep used in BDT, the Bayesian sweep evaluates previous performance and gradually converges to the best model. For the model, instead of having each layer contain 32 to 2048 neurons with steps of multiples of 2, the model expands the units in layers 1 to 3, allowing it to project into a higher dimension to capture corrections. After each layer, the model uses a ReLU activation function for nonlinearity and

a dropout rate of 10%-50% to prevent overtraining. Compression following the expansion forces the abstract features learned in the middle to condense into the most important indicators of a dark photon event. Finally, a sigmoid activation function reduces each event to a single score between 0 and 1, representing the probability that it is a signal event. To address underfitting and overfitting, an early stopping criterion of 5 epochs was implemented: training stops if the model does not improve for the next 5 epochs.

Table 11. Model of the Deep Neural Network (DNN) Classifier

Layer / Component	Configuration
Input Layer	Input (shape=N)
Expansion Phase	Layers 1–3 (up to 2048)
Activation	ReLU
Regularization	Dropout (5%–25%)
Compression Phase	Layers 4–5 (contracting)
Output Layer	Sigmoid (1 unit)
Loss Function	Binary Cross-Entropy
Sample Weights	<code>w_train</code> / <code>w_test</code>
Early Stopping	Patience = 5

Classifier performance and working point. Similar to the BDT evaluation, the performance of the best NN model is assessed by iterating over the network’s output score and calculating the simple significance ($S/\sqrt{B}$). The best run is determined by finding the threshold that maximizes this significance while retaining sufficient signal.

The nominal working point is chosen at NN score ≥ 0.03 , which yields an expected maximum significance of $S/\sqrt{B} = 3.052$ with a signal acceptance of approximately 64.04%, as shown in Fig. 23. While the maximum significance achieved by the NN is slightly lower than that of the BDT (3.052 versus 3.199),

Table 12. Hyperparameter search ranges used in the neural network Bayesian sweep and the best-performing configuration.

Hyperparameter	Scan Range / Distribution	Best Run
Learning rate	$\log_uniform(10^{-5}, 10^{-2})$	0.003883
Batch size	{128, 256, 512, 768, 1024}	1024
Dropout fraction	$uniform(0.1, 0.5)$	0.1677
Layer 1 width	{64, 128, 192, 256}	128
Layer 2 width	{128, 192, 256, 384, 512}	512
Layer 3 width	{512, 768, 1024, 1536, 2048}	512
Layer 4 width	{128, 256, 384, 512}	128
Layer 5 width	{32, 64, 96, 128}	128
Training epochs	Fixed (50)	50
Best ROC AUC	0.901	
Signal Acceptance	64.04%	
NN Score cut	0.03	
Maximum $S/\sqrt{B}$	3.052	

the neural network retains a notably higher signal acceptance (64.04% compared to 54.05%).

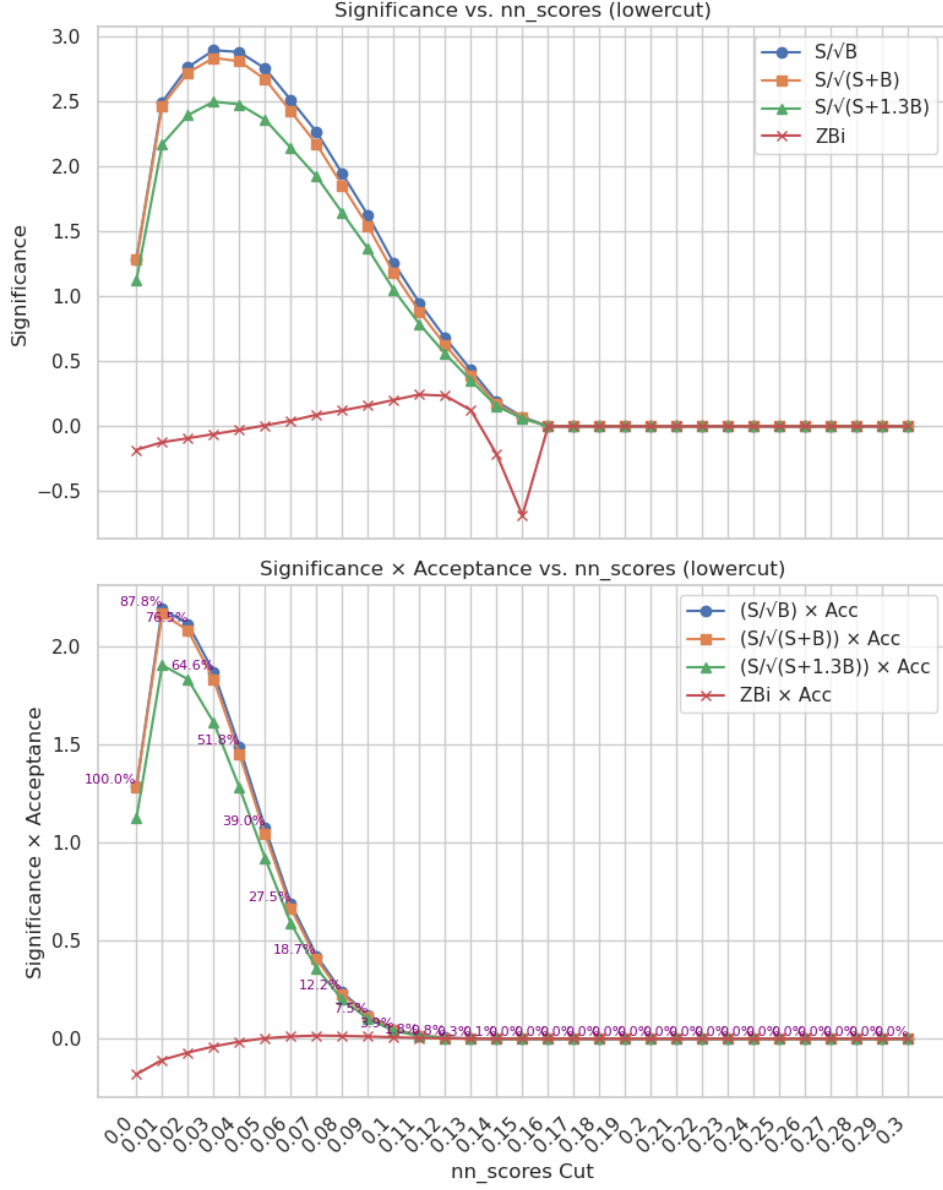


Figure 23. Significance versus NN score threshold. The maximum simple significance ($S/\sqrt{B} = 3.052$) is achieved at NN score ≥ 0.03 , corresponding to a signal acceptance of 64.04%.

3.6 Conclusions and Outlook

Table 13. Comparison of the four methods with their performance in observed and expected limit, significance, and significance x acceptance

Nominal Analysis	Optimized Selection	BDT	DNN	
observed limit	1.16%	—	—	—
expected limit	$(0.92^{+0.46}_{-0.28} \%)$	0.88%	—	—
significance	—	—	—	—
significance x acceptance	—	—	—	—

APPENDIX A

DARK PHOTON: DATA AND MC SAMPLES

Here are the tables of Data and MC Samples:

Table A.14. List of signal MCd samples used in this analysis.

mc23_13p6TeV.602403.PyPy8EG_VBF_H125yyd.deriv.DAOD_PHYS.e8523_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.602913.PyPy8EG_qqZH_H125yyd.deriv.DAOD_PHYS.e8523_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.602914.PyPy8EG_WpH_H125yyd.deriv.DAOD_PHYS.e8523_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.602915.PyPy8EG_WmH_H125yyd.deriv.DAOD_PHYS.e8523_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.602402.PyPy8EG_ggH_H125yyd.deriv.DAOD_PHYS.e8537_e8528_s4159_s4114_r15224_r15225_p6320

Table A.15. List of $\gamma jets$ MC23d and MC23e samples used in this analysis.

MC23d
mc23_13p6TeV.801663.Py8_gammajet_direct_DP8_17_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.801664.Py8_gammajet_direct_DP17_35_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.801665.Py8_gammajet_direct_DP35_50_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.801666.Py8_gammajet_direct_DP50_70_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.801667.Py8_gammajet_direct_DP70_140_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.801668.Py8_gammajet_direct_DP140_280_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.801669.Py8_gammajet_direct_DP280_500_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.801670.Py8_gammajet_direct_DP500_800_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.801671.Py8_gammajet_direct_DP800_1000_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.801672.Py8_gammajet_direct_DP1000_1500_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.801673.Py8_gammajet_direct_DP1500_2000_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.801674.Py8_gammajet_direct_DP2000_2500_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.801675.Py8_gammajet_direct_DP2500_3000_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.801676.Py8_gammajet_direct_DP3000_inf_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
MC23e
mc23_13p6TeV.801663.Py8_gammajet_direct_DP8_17_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.801664.Py8_gammajet_direct_DP17_35_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.801665.Py8_gammajet_direct_DP35_50_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.801666.Py8_gammajet_direct_DP50_70_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.801667.Py8_gammajet_direct_DP70_140_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.801668.Py8_gammajet_direct_DP140_280_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.801669.Py8_gammajet_direct_DP280_500_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.801670.Py8_gammajet_direct_DP500_800_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.801671.Py8_gammajet_direct_DP800_1000_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.801672.Py8_gammajet_direct_DP1000_1500_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.801673.Py8_gammajet_direct_DP1500_2000_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.801674.Py8_gammajet_direct_DP2000_2500_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.801675.Py8_gammajet_direct_DP2500_3000_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.801676.Py8_gammajet_direct_DP3000_inf_FullSW.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697

Table A.16. List of $W\gamma/Z\gamma$ MC23d and MC23e samples used in this analysis

MC23d
mc23_13p6TeV.700770.Sh_2214_eegamma.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.700771.Sh_2214_mumugamma.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.700772.Sh_2214_tautaugamma.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.700773.Sh_2214_enugamma.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.700774.Sh_2214_munugamma.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.700775.Sh_2214_tanugamma.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
mc23_13p6TeV.700776.Sh_2214_nunugamma.deriv.DAOD_PHYS.e8514_e8528_s4159_s4114_r15224_r15225_p6697
MC23e
mc23_13p6TeV.700770.Sh_2214_eegamma.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.700771.Sh_2214_mumugamma.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.700772.Sh_2214_tautaugamma.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.700773.Sh_2214_enugamma.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.700774.Sh_2214_munugamma.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.700775.Sh_2214_tanugamma.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697
mc23_13p6TeV.700776.Sh_2214_nunugamma.deriv.DAOD_PHYS.e8514_e8528_s4369_s4370_r16083_r15970_p6697

APPENDIX B

IRIS-HEP FELLOWSHIP PROJECT: ANALYTICAL TRACKING VALIDATION

B.1 Overview of the Project

B.1.1 Motivation and Goals.

B.1.2 A Common Tracking Software (ACTS) framework and Open Data Detector (ODD) Geometry.

B.1.3 Python Tracking Resolution Calculator.

B.2 Methodology

B.2.1 Analytical Resolution Formulation. The analytical model computes the uncertainties of the five standard ACTS track parameters: $(d_0, z_0, \phi, \theta, q/p)$. The covariance matrix is obtained from closed-form expressions derived from linearized track fits in the presence of Gaussian measurement errors and small-angle multiple scattering.

B.2.2 Measurement and Multiple Scattering Terms. The measurement term scales according to the intrinsic hit resolutions and the lever arm of the detector. It typically dominates at high momenta where multiple scattering is negligible.

The multiple scattering contribution follows the Highland approximation,

$$\theta_{\text{ms}} \simeq \frac{13.6 \text{ MeV}}{\beta p} \sqrt{\frac{x}{X_0}} \left[1 + 0.038 \ln \left(\frac{x}{X_0} \right) \right],$$

where x/X_0 is the material thickness. This term becomes dominant at low p_T and is sensitive to the accuracy of the material map.

One recurring challenge observed during the project was interpreting the material distribution in the ODD geometry. Differences between the analytical material assumptions and the more detailed ACTS material description were found to contribute significantly to discrepancies in $\sigma(d_0)$ and $\sigma(z_0)$ at low p_T .

B.2.3 Matrix Representation of Track Parameters.

Track resolutions are determined by inverting the normal-equation matrix associated with the linearized track model. The calculator constructs both the design matrix and the noise matrix using detector layer positions and resolutions, then derives the covariance matrix via:

$$\mathbf{C} = (\mathbf{A}^\top \mathbf{W} \mathbf{A})^{-1},$$

where $\mathbf{W}$ contains both measurement and scattering weights. This framework allows the analytical model to remain computationally light while still capturing key geometric dependencies.

B.3 ACTS Configuration and Simulation Setup

The ACTS validation used the full ODD geometry with a $B = 2$ T magnetic field and default digitization and reconstruction settings. Tracks were generated at fixed p_T values ranging from 1 GeV to 200 GeV and at several representative pseudorapidity values. The resulting reconstructed track parameters and covariance matrices were extracted from `tracksummary.ckf.root` and converted into flat tables for comparison.

This workflow mirrors typical detector performance studies in ATLAS and ensures that the results incorporate realistic navigation, material interactions, and Kalman filter behavior.

B.4 Comparison Strategy

For each (p_T, η) point, the analytical prediction for each parameter's resolution was compared with the RMS width (or Gaussian core width) of the ACTS simulation. The comparison was performed for:

$$- \sigma(d_0)$$

- $\sigma(z_0)$
- $\sigma(\phi)$
- $\sigma(\theta)$
- $\sigma(p_T)/p_T$

Residual fractional differences were computed to highlight systematic trends, and discrepancies were traced back to underlying assumptions such as: material description, hit resolution mapping, and the treatment of scattering correlations within the Kalman filter.

B.5 Results and Discussion

B.5.1 Resolution vs. p_T and η . Across most p_T values, the analytical model and ACTS simulation show good agreement in the high-momentum regime, where measurement errors dominate. The p_T dependence follows the expected scaling $\sigma \propto 1/p_T$ for the momentum resolution and constant behavior for angular uncertainties.

At central $\eta \approx 0$, the agreement in $\sigma(d_0)$ and $\sigma(z_0)$ is excellent above approximately 20 GeV. However, at low p_T , ACTS resolutions rise more steeply than predicted, reflecting stronger real-material multiple scattering effects.

B.5.2 Discrepancies and Model Validation. The largest discrepancies appear in regions with significant material interactions, particularly the endcap regions and low-momentum regimes. These differences were traced to:

- Simplified analytical treatment of the ODD material map,
- Non-uniform detector layer spacing in the endcaps,

- Additional scattering terms handled automatically by the ACTS Kalman filter but absent in the analytical model,
- Slight differences in hit resolution interpretation between ACTS and the analytical tool.

Despite these differences, the analytical model captures the overall behavior and correctly predicts the scale of the resolutions.

B.5.3 Implications for Fast Performance Estimation. The study confirms that analytical resolution models are highly effective for quick performance estimates in the central detector region and in the high-momentum regime. However, caution is required when applying these models in low- p_T or high- η regions where realistic material and scattering effects dominate.

This validation provides meaningful guidance for future detector optimization workflows, and it highlights the importance of maintaining consistency between analytical assumptions and full simulation frameworks such as ACTS.

B.6 Conclusions

B.6.1 Summary of Findings.

B.6.2 Cross-Project Insights.

B.6.3 Future Work.

B.7 Supplementary Material

B.7.1 ACTS and Traccc Configuration Files.

B.7.2 Code Snippets and YAML Settings.

B.7.3 Additional Figures.

REFERENCES CITED

- [1] ATLAS Collaboration, G. Aad et al., *Combined Measurement of the Higgs Boson Mass from the $H \rightarrow \gamma\gamma$ and $H \rightarrow ZZ^* \rightarrow 4\ell$ Decay Channels with the ATLAS Detector Using $s=7, 8$, and 13 TeV pp Collision Data*, Phys. Rev. Lett. **131** (2023) no. 25, 251802, [arXiv:2308.04775 \[hep-ex\]](#).
- [2] Wikimedia Commons contributors, *Standard Model of Elementary Particles*, https://en.wikipedia.org/wiki/File:Standard_Model_of_Elementary_Particles.svg, 2018. Accessed: December 10, 2024.
- [3] D. J. Griffiths, *Introduction to Elementary Particles*. Wiley-VCH, Weinheim, 2 ed., 2008.
- [4] M. Thomson, *Modern Particle Physics*. Cambridge University Press, Cambridge, 2013.
- [5] P. D. Group, *Review of Particle Physics*, Prog. Theor. Exp. Phys. **2024** (2024) no. 8, 083C01. <https://pdglive.lbl.gov>.
- [6] LHC Higgs Cross Section Working Group Collaboration, D. de Florian et al., *Handbook of LHC Higgs Cross Sections: 4. Deciphering the Nature of the Higgs Sector*, arXiv preprint (2019) , [arXiv:1901.02233 \[hep-ph\]](#). <https://arxiv.org/abs/1901.02233>.
- [7] P. Collaboration, *Planck 2018 results. VI. Cosmological parameters*, Astronomy & Astrophysics **641** (2020) A6, [arXiv:1807.06209 \[astro-ph.CO\]](#).
- [8] V. C. Rubin, W. K. Ford, and N. Thonnard, *Rotational properties of 21 Sc galaxies*, The Astrophysical Journal **238** (1980) 471.
- [9] D. Clowe, M. Bradac, A. H. Gonzalez, et al., *A Direct Empirical Proof of the Existence of Dark Matter*, The Astrophysical Journal Letters **648** (2006) L109–L113, [astro-ph/0608407](#).
- [10] S. Collaboration, *Direct Evidence for Neutrino Flavor Transformation*, Physical Review Letters **89** (2002) 011301, [nucl-ex/0204008](#).
- [11] S.-K. Collaboration, *Evidence for Oscillation of Atmospheric Neutrinos*, Physical Review Letters **81** (1998) 1562, [hep-ex/9807003](#).
- [12] A. Sakharov, *Violation of CP invariance, C asymmetry, and baryon asymmetry of the universe*, JETP Letters **5** (1967) 24.

- [13] A. Collaboration, *The ATLAS Experiment at the CERN Large Hadron Collider*, Journal of Instrumentation **3** (2008) S08003.
- [14] C. A. Groups, *The LHC Run 3 performance and plans*, CERN Yellow Reports (2022) .
- [15] A. Collaboration, *ATLAS Run 3 Physics and Detector Performance*, ATL-PHYS-PUB (2022) .
- [16] G. Apollinari et al., *High-Luminosity Large Hadron Collider (HL-LHC): Technical Design Report*, CERN Yellow Reports (2017) .
- [17] A. Collaboration, *ATLAS Inner Tracker Phase-II Upgrade Technical Design Report*, CERN-LHCC (2017) no. 2017-021, .
- [18] ATLAS Collaboration, G. Aad et al., *The ATLAS Experiment at the CERN Large Hadron Collider*, JINST **3** (2008) S08003.
- [19] E. Mobs, *The CERN accelerator complex*, <https://cds.cern.ch/record/2684277>, July, 2019. General Photo.
- [20] ATLAS Collaboration, *Luminosity Public Results for LHC Run 3*, <https://twiki.cern.ch/twiki/bin/view/AtlasPublic/LuminosityPublicResultsRun3>, 2023. Accessed: 2026-01.
- [21] ATLAS Collaboration, S. Mehlhase, *ATLAS detector slice (and particle visualisations)*, tech. rep., 2021. <https://cds.cern.ch/record/2770815>.
- [22] ATLAS Collaboration, *The ATLAS trigger system for LHC Run 3 and trigger performance in 2022*, JINST **19** (2024) no. 06, P06029, arXiv:2401.06630 [hep-ex]. arXiv:2401.06630 [hep-ex].
- [23] G. Apollinari et al., *High-Luminosity Large Hadron Collider (HL-LHC): Technical Design Report*. CERN, 2020.
- [24] A. Collaboration, *ATLAS Phase-II Upgrade Technical Design Report*, CERN-LHCC (2017) no. 2017-021, .
- [25] ATLAS Collaboration, *Technical Design Report for the ATLAS Inner Tracker Strip Detector*, tech. rep., CERN, Apr, 2017. <https://cds.cern.ch/record/2257755>.
- [26] ATLAS Collaboration, *Technical Design Report for the ATLAS Inner Tracker Pixel Detector*, tech. rep., CERN, Sep, 2017. <https://cds.cern.ch/record/2285585>.

- [27] ATLAS Collaboration, *Technical Design Report for the Phase-II Upgrade of the ATLAS TDAQ System*, tech. rep., CERN, Sep, 2017.
<https://cds.cern.ch/record/2285584>.
- [28] ATLAS Collaboration, *Technical Design Report for the Phase-II Upgrade of the ATLAS Trigger and Data Acquisition System - Event Filter Tracking Amendment*, tech. rep., CERN, Mar, 2022. <https://cds.cern.ch/record/2802799>.
- [29] V. Boisvert et al., *ATLAS TDAQ Phase-II Upgrade: EF Tracking User Requirements Document*, tech. rep., CERN, Geneva, 2023.
<https://edms.cern.ch/document/2960991/>.
- [30] F. Winklmeier, *Phase-II Event Filter requirements*, tech. rep., CERN, Geneva, 2023. <https://edms.cern.ch/document/3009033/1>.
- [31] ATLAS Collaboration, *Trigger Studies with EF Tracking at the HL-LHC*, tech. rep., CERN, Geneva, 2024. <https://cds.cern.ch/record/2916764>.
- [32] V. Boisvert and S. Majewski, *ATLAS TDAQ Phase-II Upgrade: EF Tracking Technology Choice Criteria Document*, tech. rep., CERN, Geneva, 2024.
<https://edms.cern.ch/document/3022957/>.
- [33] S. George, *Report of the Reviews of the EF Tracking URD and TCC documents*, tech. rep., CERN, Geneva, 2024. <https://edms.cern.ch/document/3163250/>.
- [34] C. Gumpert et al., *ACTS: from ATLAS software towards a common track reconstruction software*, Journal of Physics: Conference Series **898** (Jan, 2017) 042011.
- [35] T. Adye et al., *ATLAS TDAQ Phase-II Upgrade: EF Tracking C-100/230 Pipeline Report*, tech. rep., CERN, Geneva, 2025.
<https://cds.cern.ch/record/2939310>.
- [36] ACTS Project, *Traccc*, Github repository, 2025.
<https://github.com/acts-project/traccc>.
- [37] ACTS Collaboration, X. Ai et al., *A Common Tracking Software Project*, Comput. Softw. Big Sci. **5** (2021) no. 1, 20, [arXiv:2106.13593](https://arxiv.org/abs/2106.13593) [physics.ins-det].
- [38] J. Fan, A. Katz, L. Randall, and M. Reece, *Dark-Disk Matter*, Phys. Rev. Lett. **110** (2013) no. 21, 211302, [arXiv:1303.1521](https://arxiv.org/abs/1303.1521) [astro-ph.CO].
- [39] M. Heikinheimo, T. Tenkanen, K. Tuominen, and V. Vaskonen, *Observational Constraints on Decoupled Hidden Sectors*, Phys. Rev. D **92** (2015) 023506, [arXiv:1504.04235](https://arxiv.org/abs/1504.04235) [astro-ph.CO].

- [40] N. Arkani-Hamed, D. P. Finkbeiner, T. R. Slatyer, and N. Weiner, *A Theory of Dark Matter*, Phys. Rev. D **79** (2009) 015014, [arXiv:0810.0713 \[hep-ph\]](#).
- [41] K. M. Zurek, *Asymmetric Dark Matter: Theories, Signatures, and Constraints*, Phys. Rept. **537** (2014) 91–121, [arXiv:1308.0338 \[hep-ph\]](#).
- [42] B. Holdom, *Two $U(1)$ ’s and Epsilon Charge Shifts*, Physics Letters B **166** (1986) no. 2, 196–198.
- [43] R. Essig et al., *Dark Sectors and New, Light, Weakly-Coupled Particles*, Journal of High Energy Physics **11** (2013) 167, [arXiv:1311.0029 \[hep-ph\]](#).
- [44] J. Beacham et al., *Physics Beyond Colliders at CERN: Beyond the Standard Model Working Group Report*, J. Phys. G **47** (2020) no. 1, 010501, [arXiv:1901.09966 \[hep-ex\]](#).
- [45] M. Fabbrichesi, E. Gabrielli, and G. Lanfranchi, *The Dark Photon*, SpringerBriefs in Physics (2021) , [arXiv:2005.01515](#).
- [46] E. Gabrielli, M. Heikinheimo, B. Mele, and M. Raidal, *Dark photons and resonant transitions of rare meson decays*, Phys. Rev. D **90** (2014) no. 5, 055032, [arXiv:1405.5191 \[hep-ph\]](#).
- [47] S. Biswas, E. Gabrielli, M. Heikinheimo, and B. Mele, *Dark-photon searches via Higgs-boson production at the LHC*, Phys. Rev. D **93** (2016) no. 9, 093011, [arXiv:1603.01377 \[hep-ph\]](#).
- [48] H. Beauchesne and C.-W. Chiang, *Observability of the Higgs boson decay to a photon and a dark photon*, Phys. Rev. D **108** (July, 2023) 015018, [arXiv:2304.04165 \[hep-ph\]](#).
- [49] H. Beauchesne and C.-W. Chiang, *Is the Decay of the Higgs Boson to a Photon and a Dark Photon Currently Observable at the LHC?*, Phys. Rev. Lett. **130** (Apr., 2023) 141801, [arXiv:2205.10976 \[hep-ph\]](#).
- [50] ATLAS Collaboration, *Search for the Higgs boson decay $H \rightarrow \gamma\gamma_d$ in the VBF production mode with the ATLAS detector*, JHEP **03** (2023) 042.
- [51] ATLAS Collaboration, *Search for dark matter in events with a photon and missing transverse momentum in pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*, JHEP **02** (2021) 226.
- [52] CMS Collaboration, *Search for a Higgs boson decaying into a photon and a massless dark photon*, JHEP **09** (2023) 104.
- [53] ATLAS Collaboration, *Combination of searches for Higgs boson decays into a photon and a dark photon using 139 fb^{-1} of pp collision data at $\sqrt{s} = 13$ TeV with the ATLAS detector*, ATLAS-CONF-2024-001 (2024) .

- [54] ATLAS Luminosity Working Group, *Luminosity for Physics*, <https://twiki.cern.ch/twiki/bin/viewauth/Atlas/LuminosityForPhysics>, 2024. Accessed: 2026-02-04.
- [55] ATLAS Collaboration, *Measurements of the photon identification efficiency with the ATLAS detector*, Eur. Phys. J. C **79** (2019) 205.
- [56] ATLAS Collaboration, *Muon reconstruction performance of the ATLAS detector*, Eur. Phys. J. C **76** (2016) 292.
- [57] ATLAS Collaboration, *Electron reconstruction and identification in the ATLAS experiment*, Eur. Phys. J. C **79** (2019) 639.
- [58] ATLAS Collaboration, *Jet energy scale and resolution measured in proton–proton collisions at 13 TeV with the ATLAS detector*, Phys. Rev. D **96** (2017) 072002.
- [59] ATLAS Collaboration, *ATLAS object definitions for physics analyses*, 2023. ATL-PHYS-PUB (object definitions public note).
- [60] ATLAS Collaboration, *Performance of missing transverse momentum reconstruction in proton–proton collisions at 13 TeV*, Eur. Phys. J. C **78** (2018) 903.
- [61] ROOT Collaboration, K. Cranmer, G. Lewis, L. Moneta, A. Shibata, and W. Verkerke, *HistFactory: A tool for creating statistical models for use with RooFit and RooStats*, tech. rep., New York U., New York, 2012. <https://cds.cern.ch/record/1456844>.
- [62] A. L. Read, *Presentation of search results: the CL_S technique*, J. Phys. G **28** (2002) 2693.
- [63] G. Cowan, K. Cranmer, E. Gross, and O. Vitells, *Asymptotic formulae for likelihood-based tests of new physics*, Eur. Phys. J. C **71** (2011) 1554, [arXiv:1007.1727 \[physics.data-an\]](#).
- [64] ATLAS Collaboration, *Combination of searches for Higgs boson decays into a photon and a massless dark photon using pp collisions at $\sqrt{s} = 13$ TeV with the ATLAS detector*, Journal of High Energy Physics **08** (2024) 153, [arXiv:2406.01656 \[hep-ex\]](#).
- [65] J. H. Friedman, *Greedy Function Approximation: A Gradient Boosting Machine*, Annals of Statistics **29** (2001) no. 5, 1189–1232.
- [66] A. Höcker et al., *TMVA – Toolkit for Multivariate Data Analysis*, PoS **ACAT** (2007) 040.

- [67] F. Pedregosa et al., *Scikit-learn: Machine Learning in Python*, Journal of Machine Learning Research **12** (2011) 2825–2830.
- [68] L. Biewald, *Experiment Tracking with Weights and Biases*, 2020.
<https://www.wandb.com/>. Software available from wandb.com.